

Mobility Management

Advanced Mobile Communication Networks



Integrated Communication Systems Group
Ilmenau University of Technology

Motivation

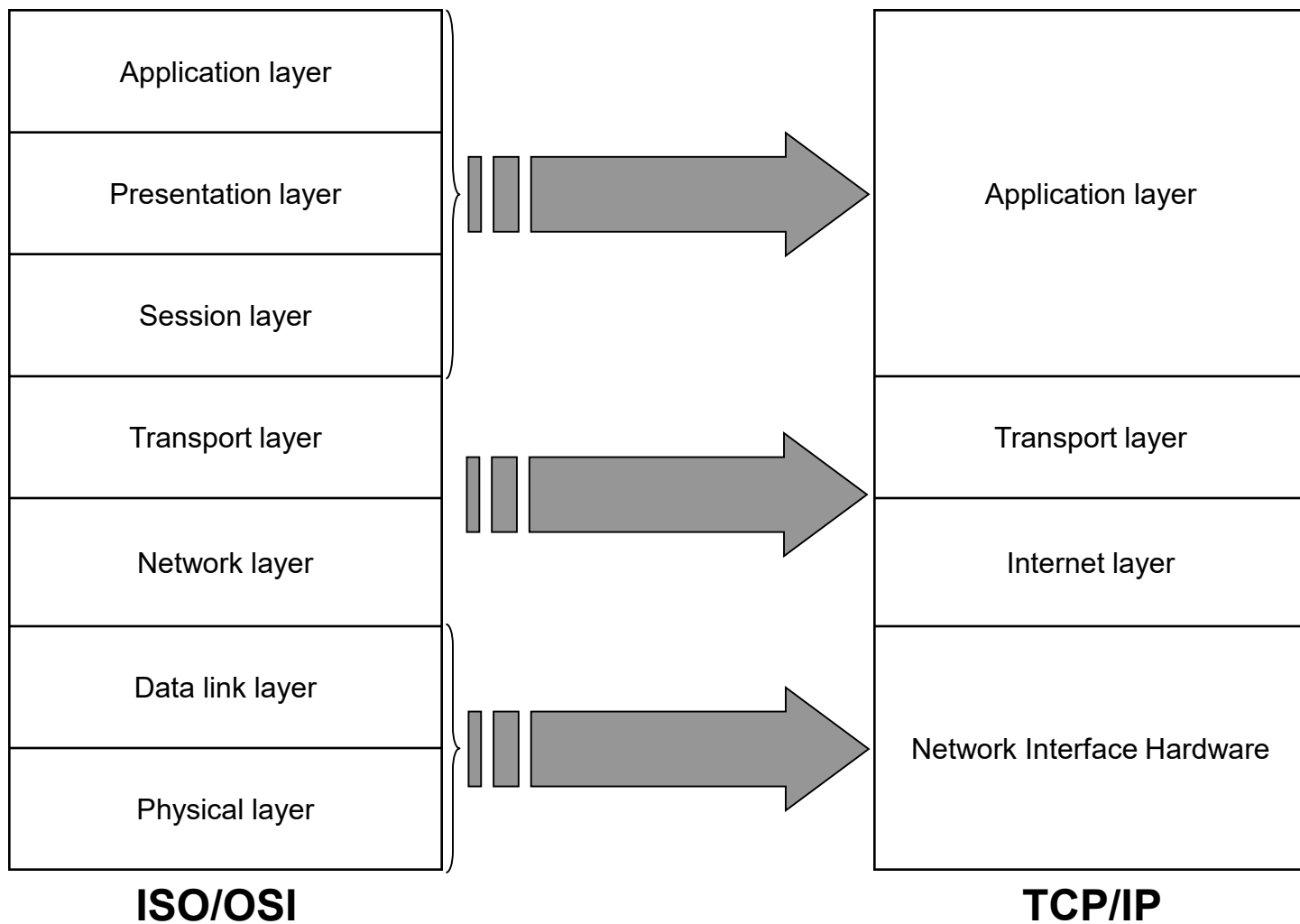
- Goals: continuous connectivity independent of location with high bandwidth, low latency, high reliability, guaranteed quality and at low cost
- Challenge: transparent support of seamless mobility (seamless handoff) between different points of access to the Internet (from changing subnets)

Mobility Management in the Reference Models

Questions to be answered:

- What are the tasks of mobility management?
- Which protocol layer should handle the mobility problem? What are the implications of selecting one or another layer?
- What are the performance criteria for handovers?
- Why can the cooperation between layers improve the handover performance?

Review of OSI and TCP/IP Reference model



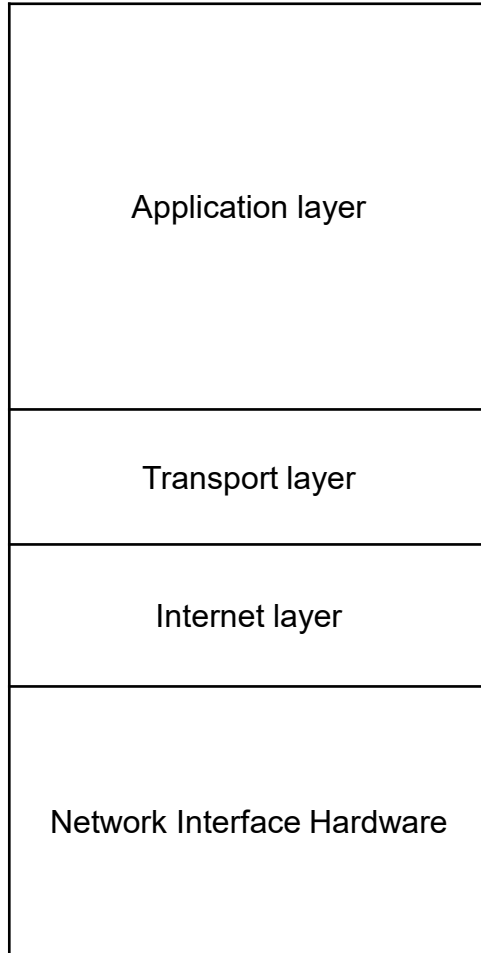
Which Layer Handles Mobility?

Application layer	Application layer mobility	Application layer
Presentation layer		
Session layer	Session layer mobility	Transport layer
Transport layer	Transport layer mobility	
Network layer	Network layer mobility	Internet layer
Data link layer	Link layer mobility	Network Interface Hardware
Physical layer	PHY layer mobility	
ISO/OSI		TCP/IP

Which Layer Handles Mobility?

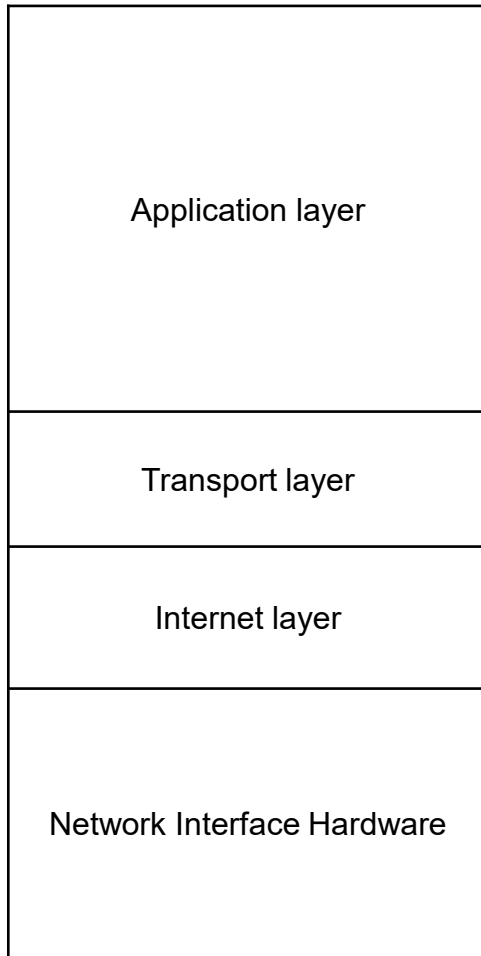
- If the old as well as the new AP belong to the **same subnet**, the problem is handled locally in the IP subnet
- **Link (and PHY) layer mobility management** is responsible for the establishment of a radio link between the Mobile Node (MN) and the new Access Point (AP)
- If the new AP belongs to a **different subnet** (i.e. uses different IP addresses), we need
- **Global mobility support** either
 - in the **network layer**, i.e. the **Internet layer** of the TCP/IP reference model,
 - in the **transport layer** or
 - in the **application layer**

Mobility Management in Different Layers, Why?



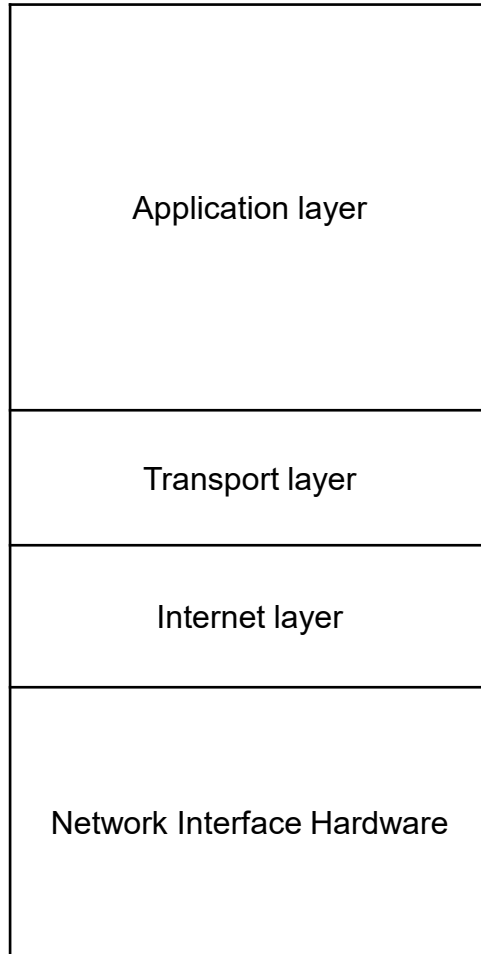
- **Link (and PHY) layer mobility** (local mobility)
 - Required when changing the point of attachment within the subnet
 - Responsible for establishment of a wireless link

Mobility Management in Different Layers, Why?



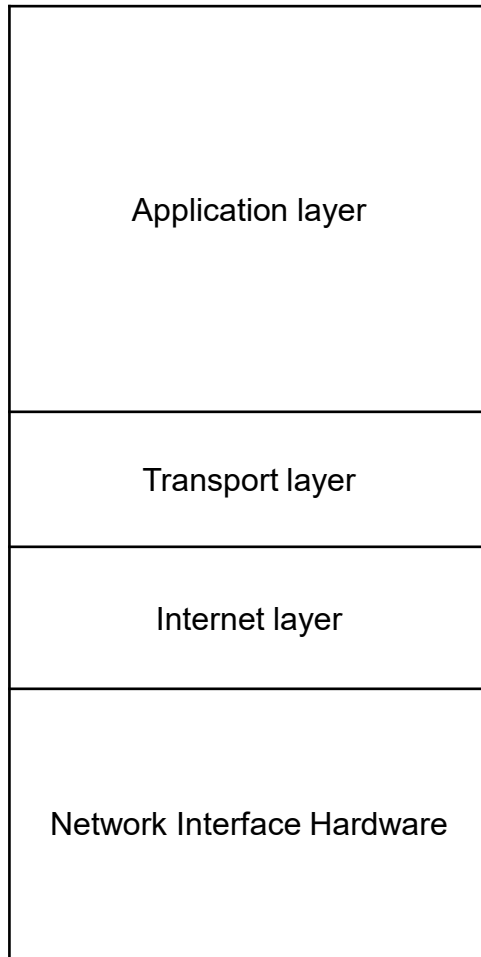
- **Network layer mobility** (global mobility)
 - Transparency to higher layer protocols, e.g. TCP and UDP
 - Applications on mobiles can further communicate with existing applications without any modifications
 - Change of network architecture is allowed

Mobility Management in Different Layers, Why?



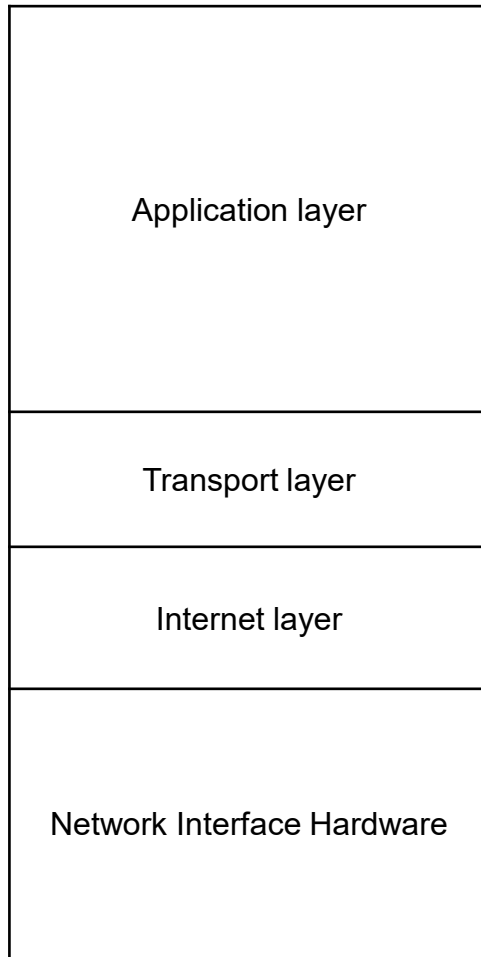
- **Transport layer mobility**
 - End-to-end mobility management while keeping the Internet infrastructure unchanged
 - End hosts take care of mobility, i.e. TCP and UDP are updated to support mobility

Mobility Management in Different Layers, Why?



- **Application layer mobility**
 - No changes to current networks
 - Example: extending IP telephony infrastructure to fulfill mobility requirements
 - Usage of SIP protocol and support for mobility by change of the mapping between a name of a user (e.g. mail address) and the IP address
- **Session layer mobility**
 - Migration of sessions between devices

Mobility Management in Different Layers, Why?



- **Hybrid layer mobility**
 - Optimization of the performance of a certain layer mobility management approach using information from other layers or the integration of solutions from several layers
 - Synchronization between layers is essential to provide a good handover performance

Tasks and criteria of mobility management

- **Tasks of mobility management**

- Find (also called paging) and address a mobile away from home
- Support “continuous” communication after change of location
- Security: avoid misuse of mechanism
- Privacy: hide location from others

- **Criteria to compare mobility protocols**

- Handover (HO) performance
 - Delay to update the route
 - Packet loss due to handover
- Protocol overhead: consumption of network resources
- Scalability: protocol overhead in large networks with large numbers of mobile nodes
- Robustness: adaptability to different network conditions and failures
- Ease of deployment: simplicity, suitability to different network scenarios

Link Layer Mobility Management (Layer 2 and PHY)

Questions to be answered:

- Which mobility problems can be handled on the link layer? Which not!
- What happens if a mobile moves from one WLAN coverage area to another? What is the main source for handover delay?
- How could this be accelerated?

Link Layer Handoff

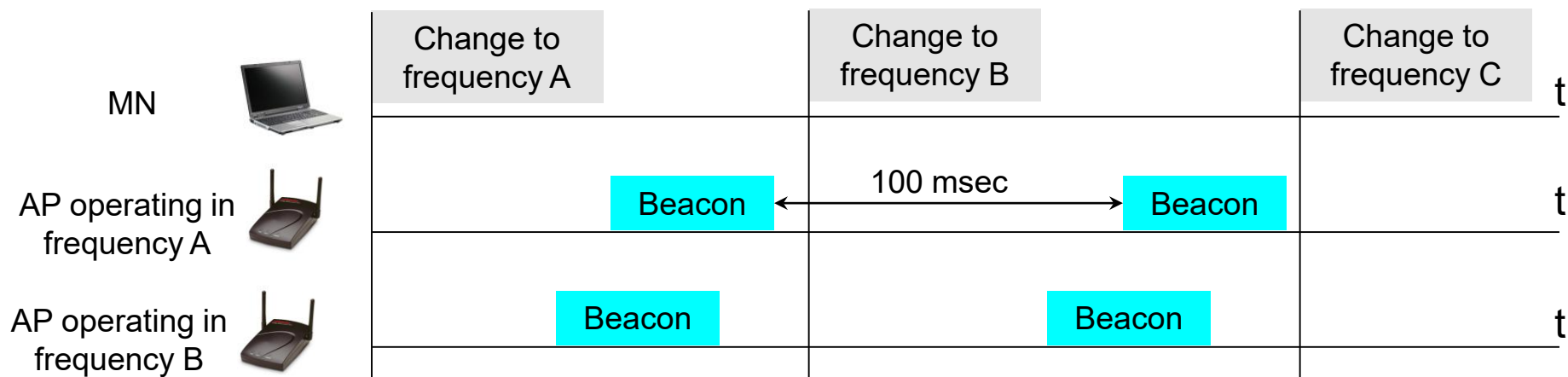
- This scenario considers handoffs between WLAN access points (APs) as an example!
 - Other communication systems, e.g. GSM, UMTS or LTE, use more sophisticated handoff mechanisms!
- Responsible for the establishment of a radio link between the MN and the AP (and connectivity within the subnet)
- The handoff comprises 4 phases
 - Recognition of the loss of the wireless connection
 - Search for and detection of a new adequate AP
 - Re-/Authentication with the new discovered AP
 - Re-/Association with the new discovered AP

Recognizing the Loss of the Wireless Connection

- Connection loss is detected based on
 - Weakness of the received signal (received signal strength) or
 - Failed frame transmissions (packet/frame error rate)
- Weakness of the received signal
 - Most frequently used approach
 - A link layer handoff is prompted, if the received signal strength goes below a certain threshold
- Failed frame transmissions
 - Slower than received-signal-strength-based approach
 - The MN first assumes a collision as a reason for failed frame transmissions; if not, radio signal fading is assumed; if not, an out of range is declared

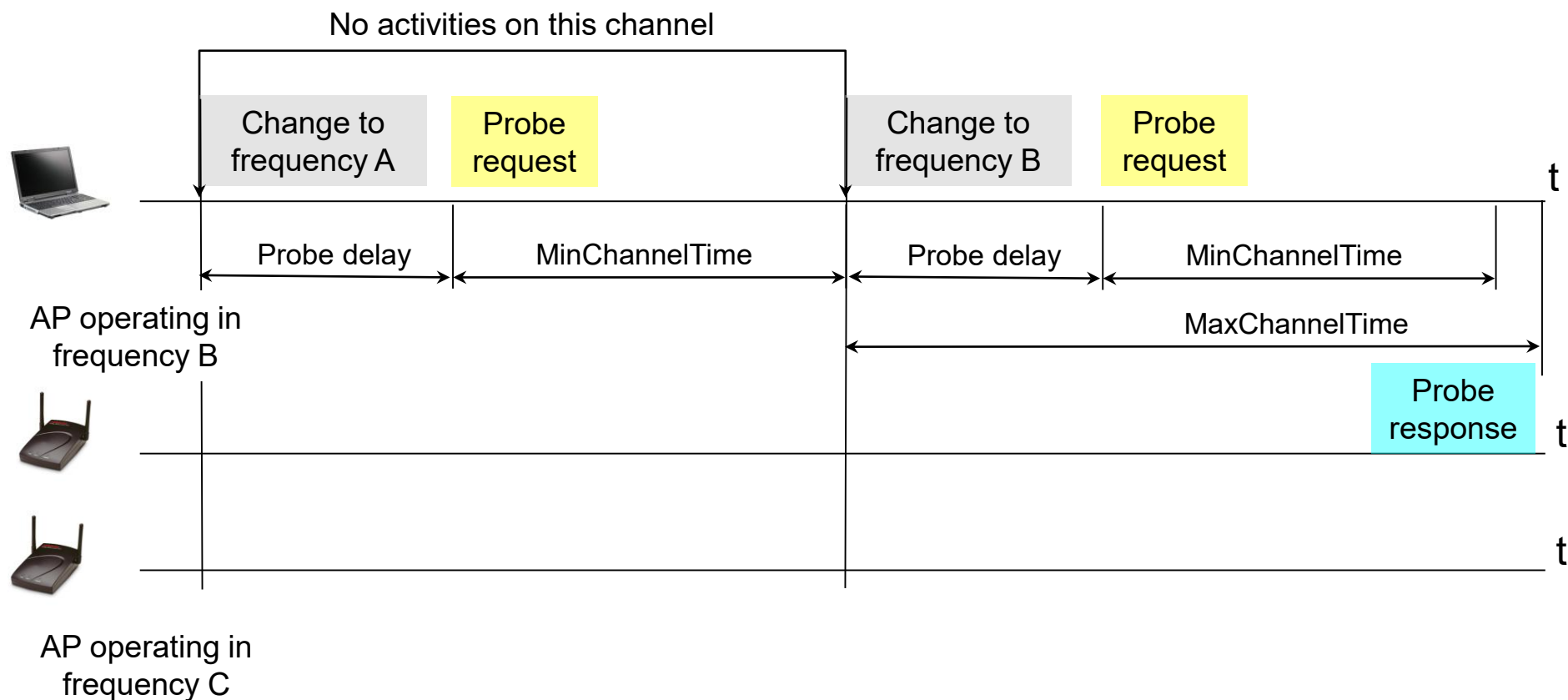
Search and Detection of a New Adequate AP

Passive scanning



Search and Detection of a New Adequate AP

Active scanning



Re-/Authentication with the New Discovered AP

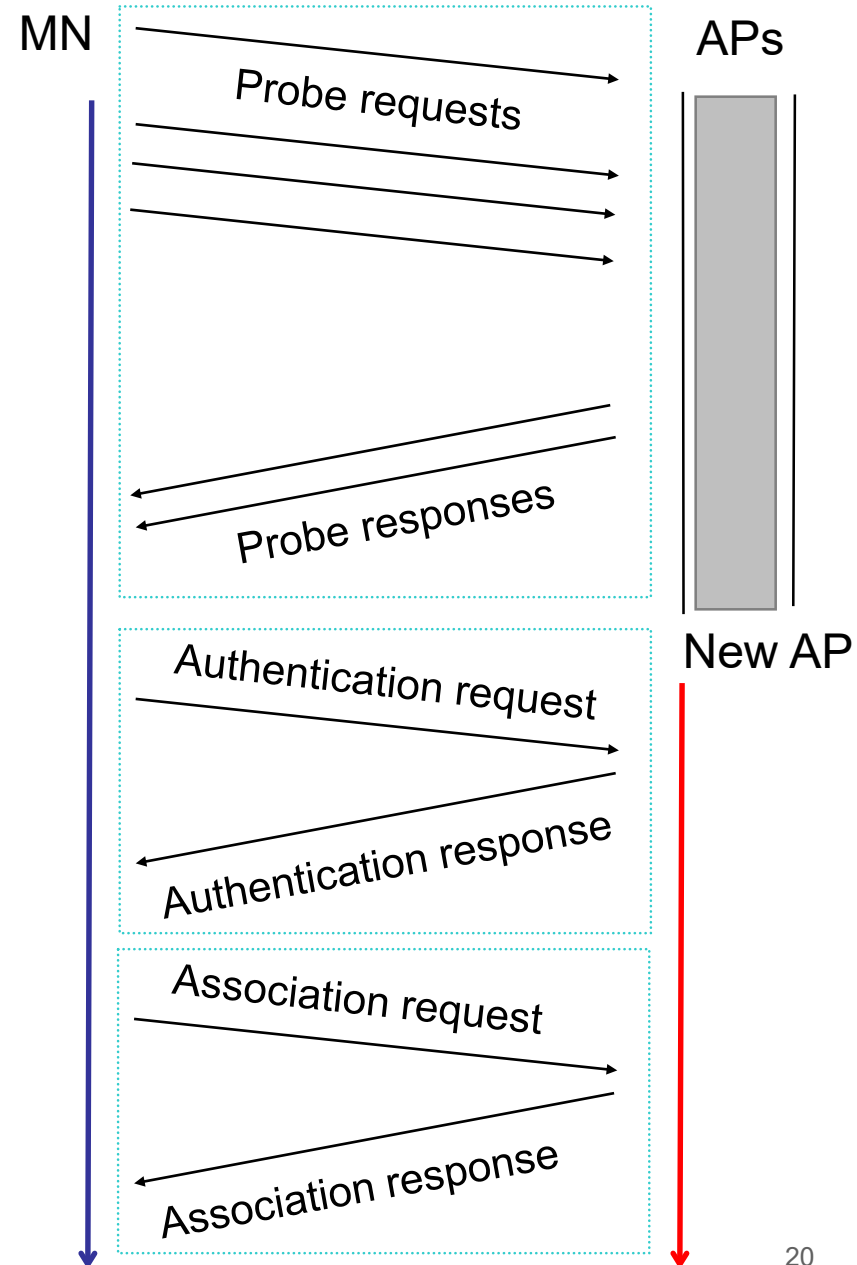
- The MN authenticates itself to the new AP
- Various authentication methods are defined for the IEEE 802.11 standard
 - Open system: null-authentication (default method)
 - Shared key authentication
 - FT authentication
 - Simultaneous Authentication of Equals (SAE)
- Exchange of authentication request & authentication response messages
- More messages can be exchanged between the authentication request & response; details depend on the authentication method

Re-/Association With the New Discovered AP

- Exchange of association request & association response messages
- After this phase, the wireless link is established

Link Layer Handoff

- Large variation for the same hardware with same configuration mainly due to stochastic behavior of radio interface
- With 802.11 scanning is the main factor for handoff latency, it accounts for about 90 % of the overall handoff delay



Issues for Improvements

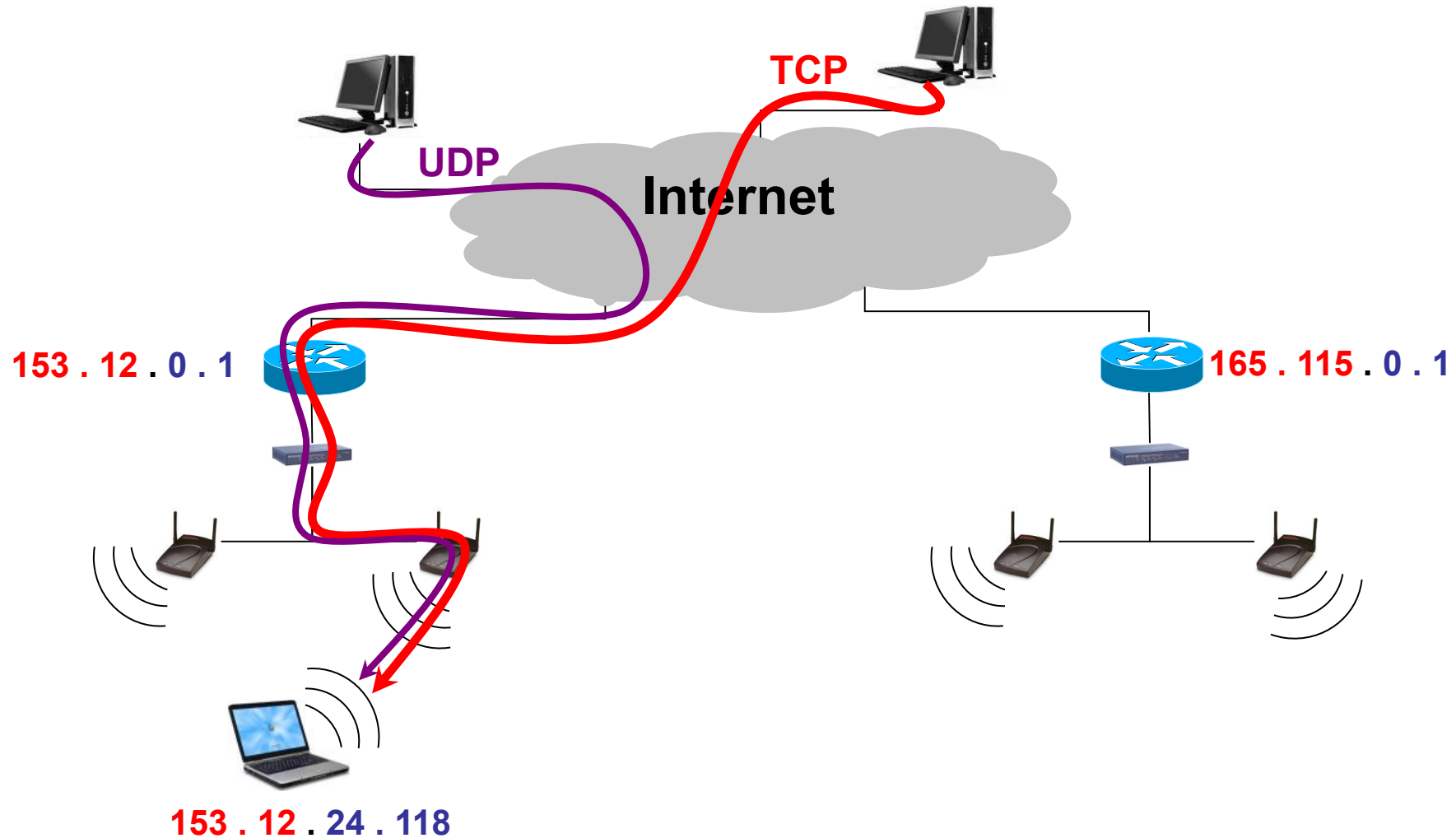
- Speeding up the link layer handoff in 802.11 requires accelerating the scanning phase
 - Periodic scanning
 - Scanning the medium while the MN is still connected to the old AP
 - When the signal strength of the current AP is decreasing, the MN switches to another frequency for a short time and scans for other available APs
 - Information about neighbors
 - Utilizing information about neighbor APs to reduce the number of channels that must be scanned
- Cellular networks apply additional techniques to accelerate the handover
 - The base station is aware of all neighbors and organizes the handover for the mobile before the current link fails (make-before-break applied in all cellular networks)
 - The mobile may also be connected to multiple base stations at the same time (soft handover/macro-diversity in UMTS)

Network Layer Mobility Management

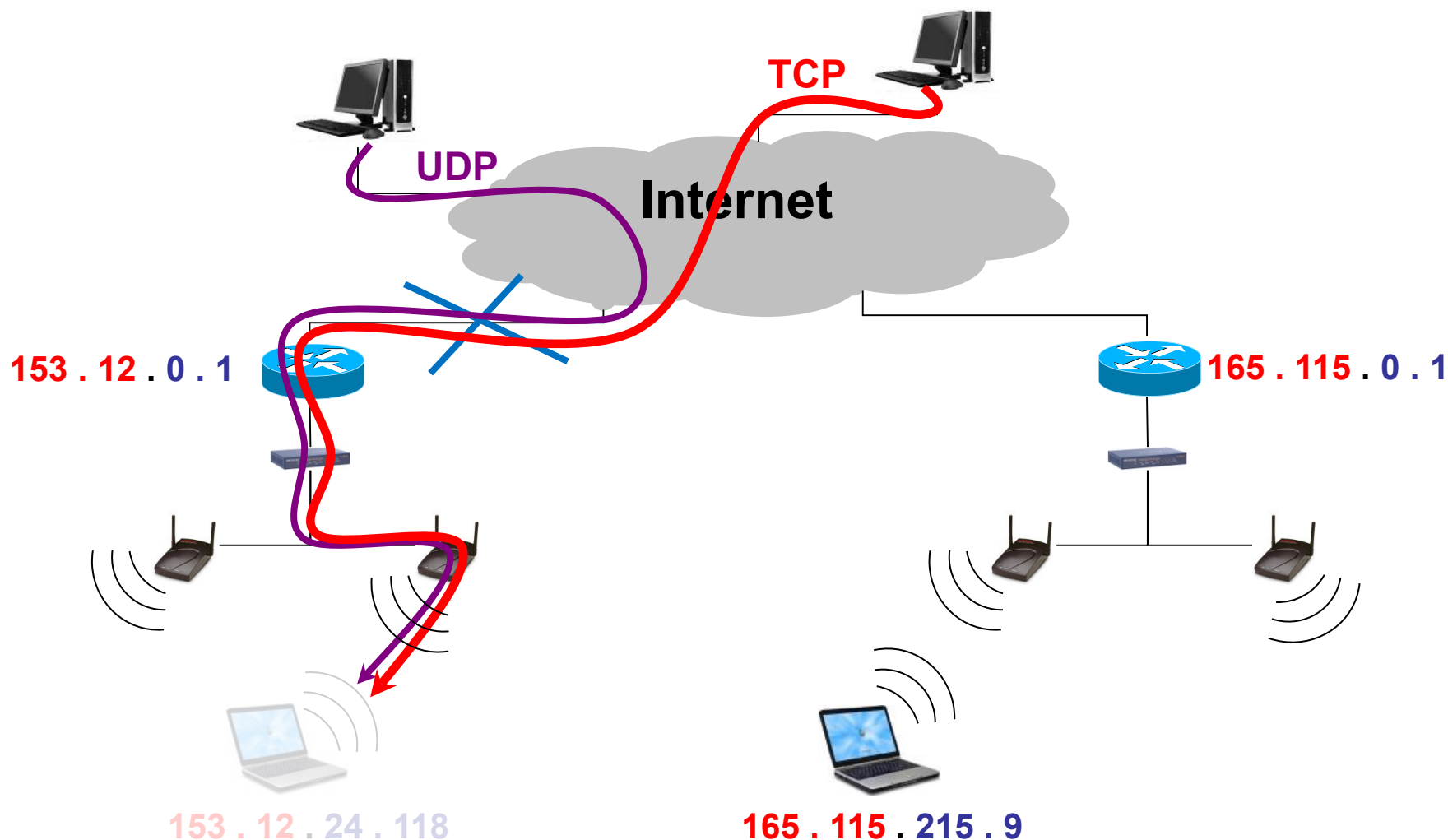
Questions to be answered:

- How can mobility be supported by the network layer?
- When does the sole update of routing tables make sense and when not? Why does it not scale?
- What is the problem with changing the IP address for mobiles due to mobility?
- What is the purpose of tunneling? What is the advantage of this over specific routes?

What is the problem of global mobility?



What is the problem of global mobility?



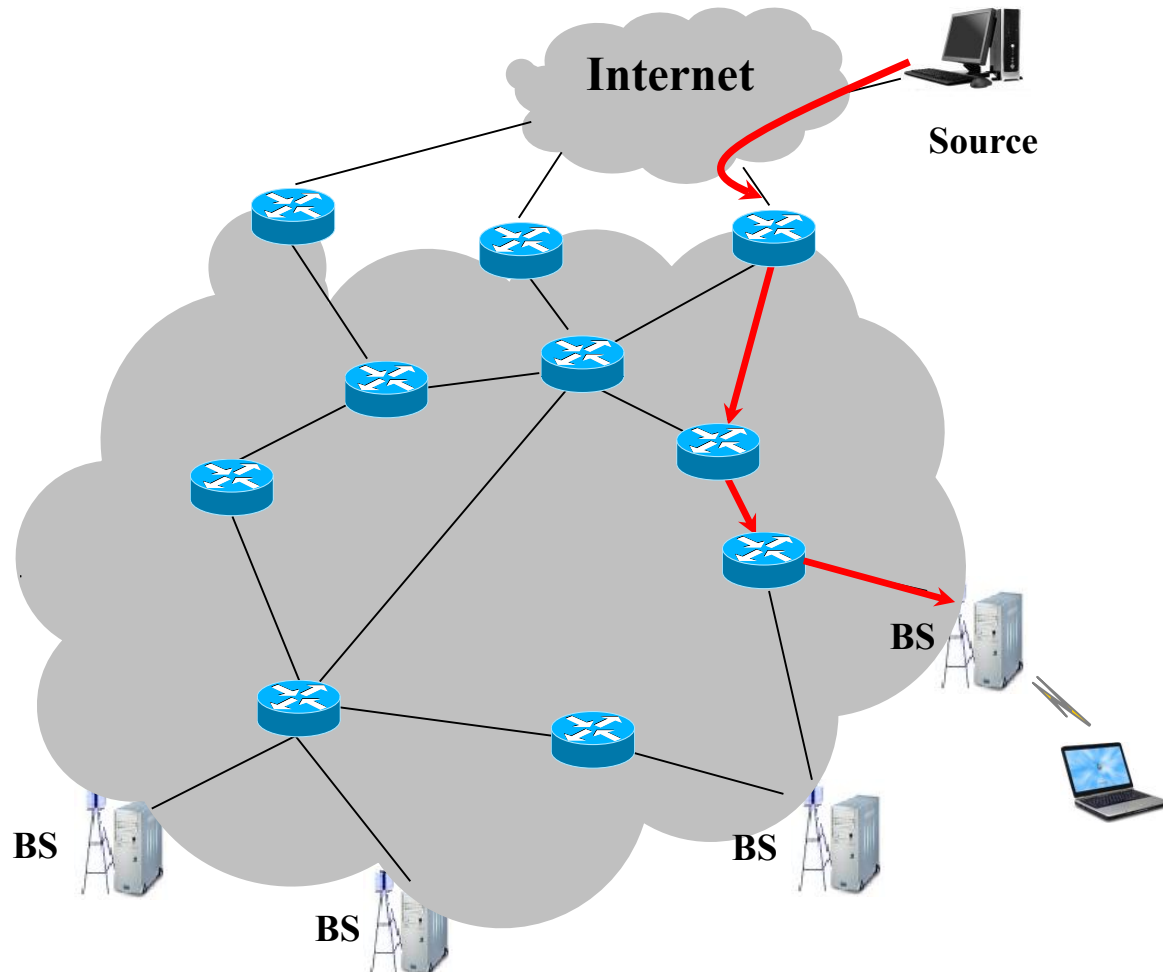
Communication disruption should be minimized or even eliminated

What is the problem and how to solve it?

- Routing
 - Forwarding is based on some destination address (a locator)
 - In IP-based networks this is the IP address on the network layer and the MAC address on the link layer
 - Change of location (due to mobility) requires change of destination address or needs an update of routing tables
- Ideas
 - Specific routes to end systems (per-host forwarding)
 - Multicast
 - Modification of the destination address
 - Tunneling

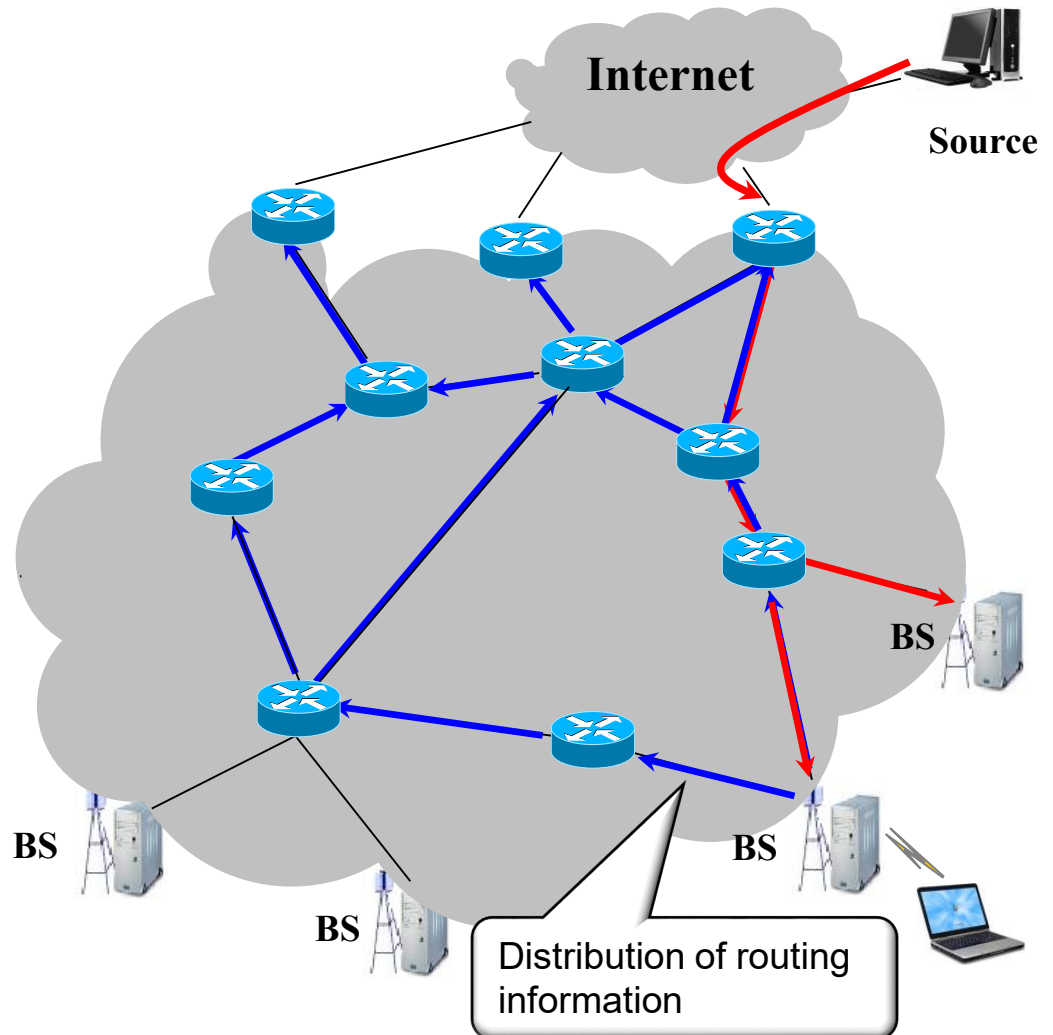
Specific routes to end systems

- Change of routing table entries to forward packets to the right destination
- Distribution of routing information for every individual mobile everywhere rather than for each subnet



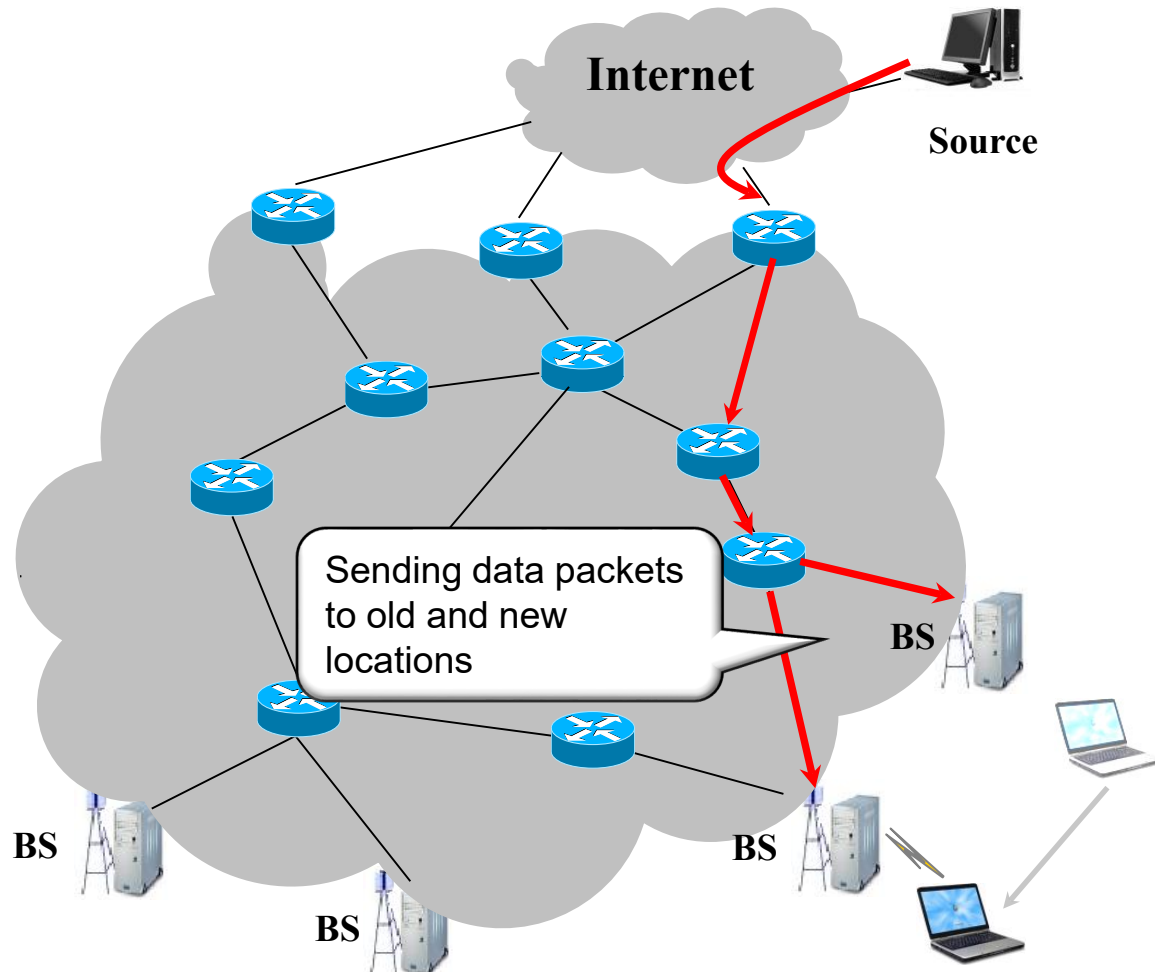
Specific routes to end systems

- Change of routing table entries to forward packets to the right destination
- Distribution of routing information for every individual mobile (rather than subnets) everywhere
- **Does not scale to a large network**
- Security problems (routing table updates)



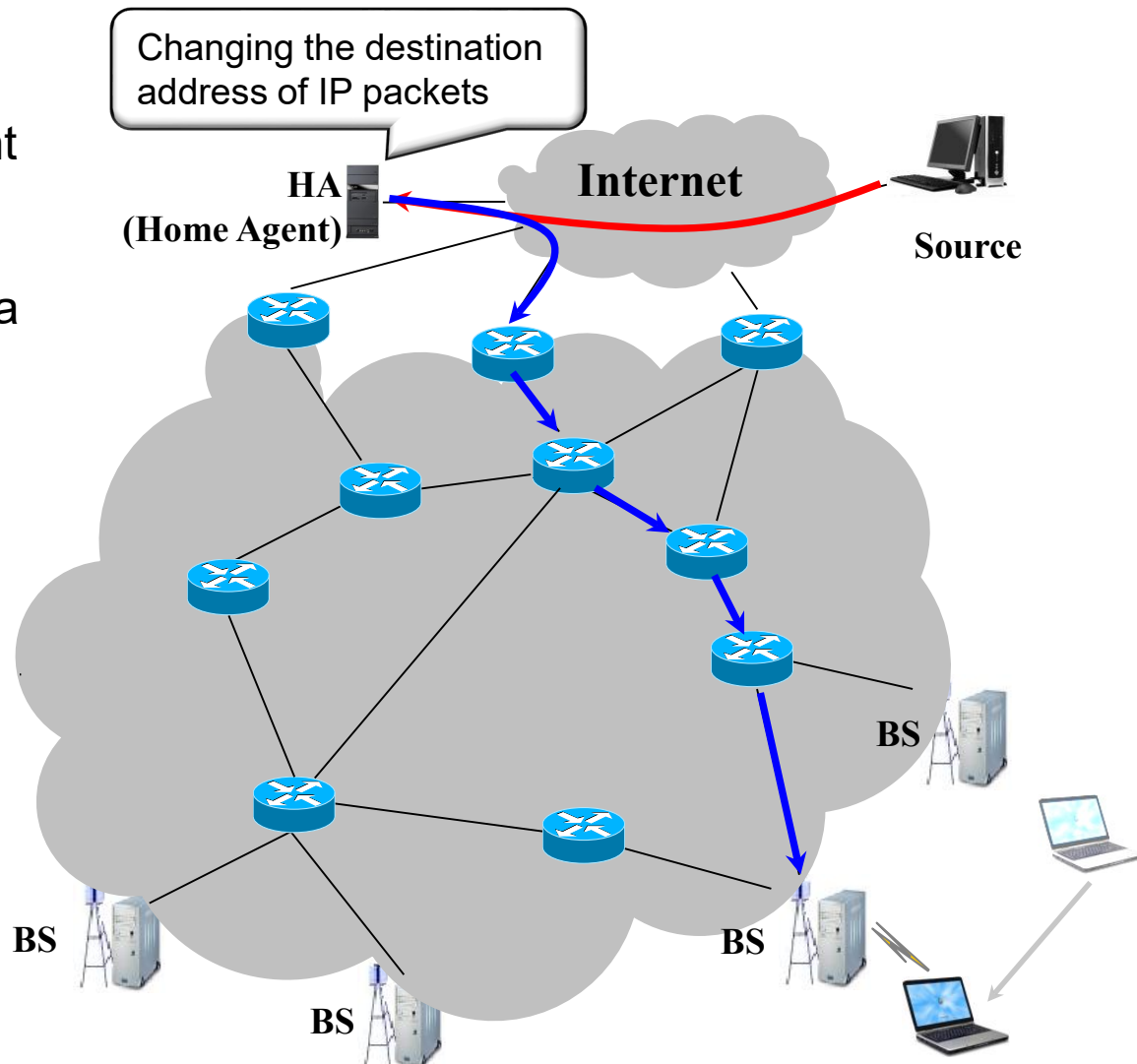
Multicast

- Building of a multicast tree
 - Security problem
 - Overhead



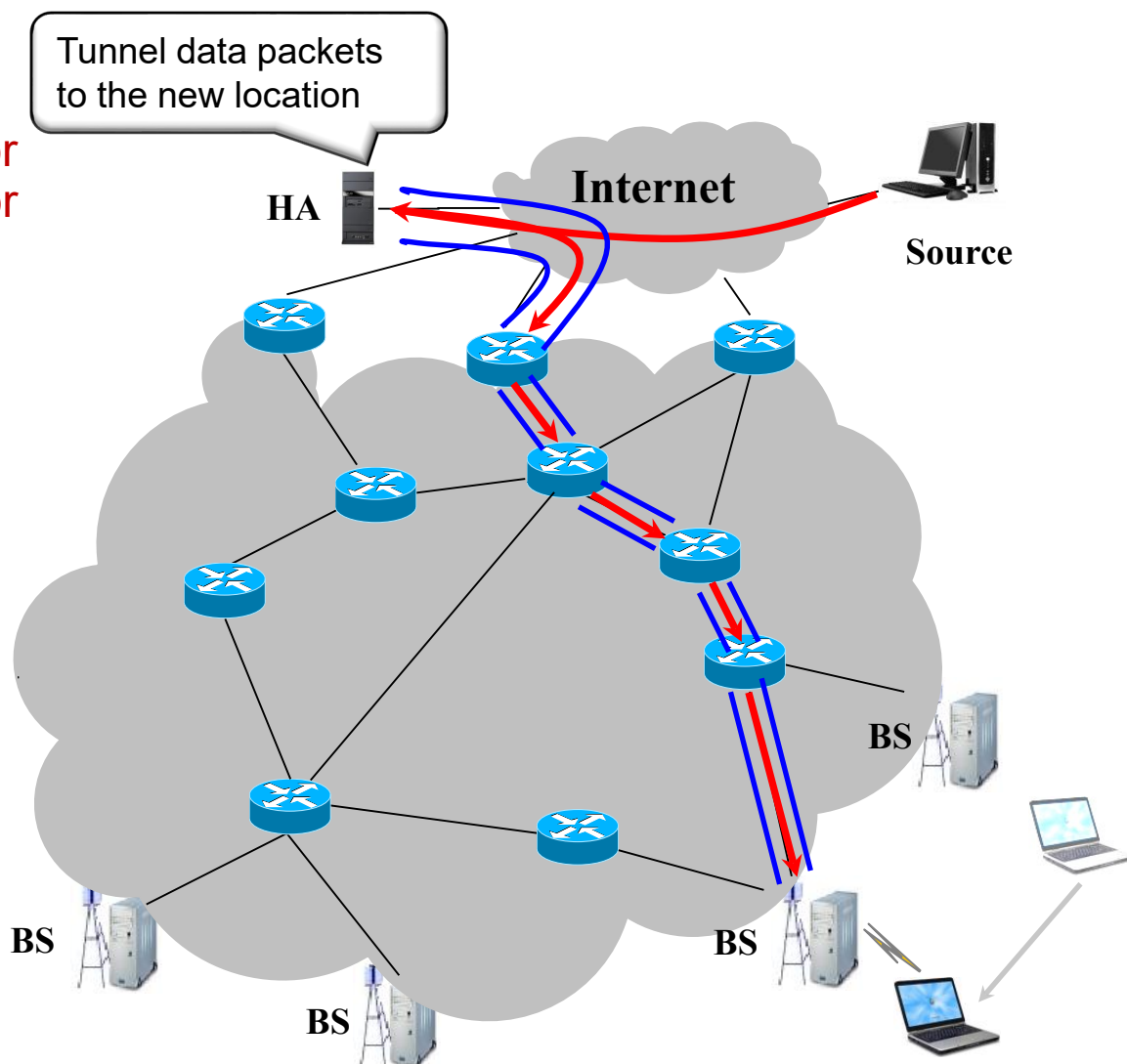
Modification of the destination address

- Adjust the host address depending on the current location
 - Problems to find a mobile host without a constant address
 - Problems with application due to change of address
 - Security problem



Tunneling

- Separation of a **terminal/user identifier** from a **topological locator**
 - Security problem
 - Location update at HA



Mobile IP

Questions to be answered:

- What is the overhead imposed on data traffic with Mobile IP?
- What is the overhead for signaling with Mobile IP?
- When is triangular routing especially harmful? Can we solve this and how?
- What are the differences between the CoA-handling in the FA vs. handling in the MN itself?
- Which optimizations can be made in order to smooth data transport after a handover?
- What is the purpose of reverse tunneling? In which case is it helpful?
- What is the purpose of DHCP? What problem does it solve?

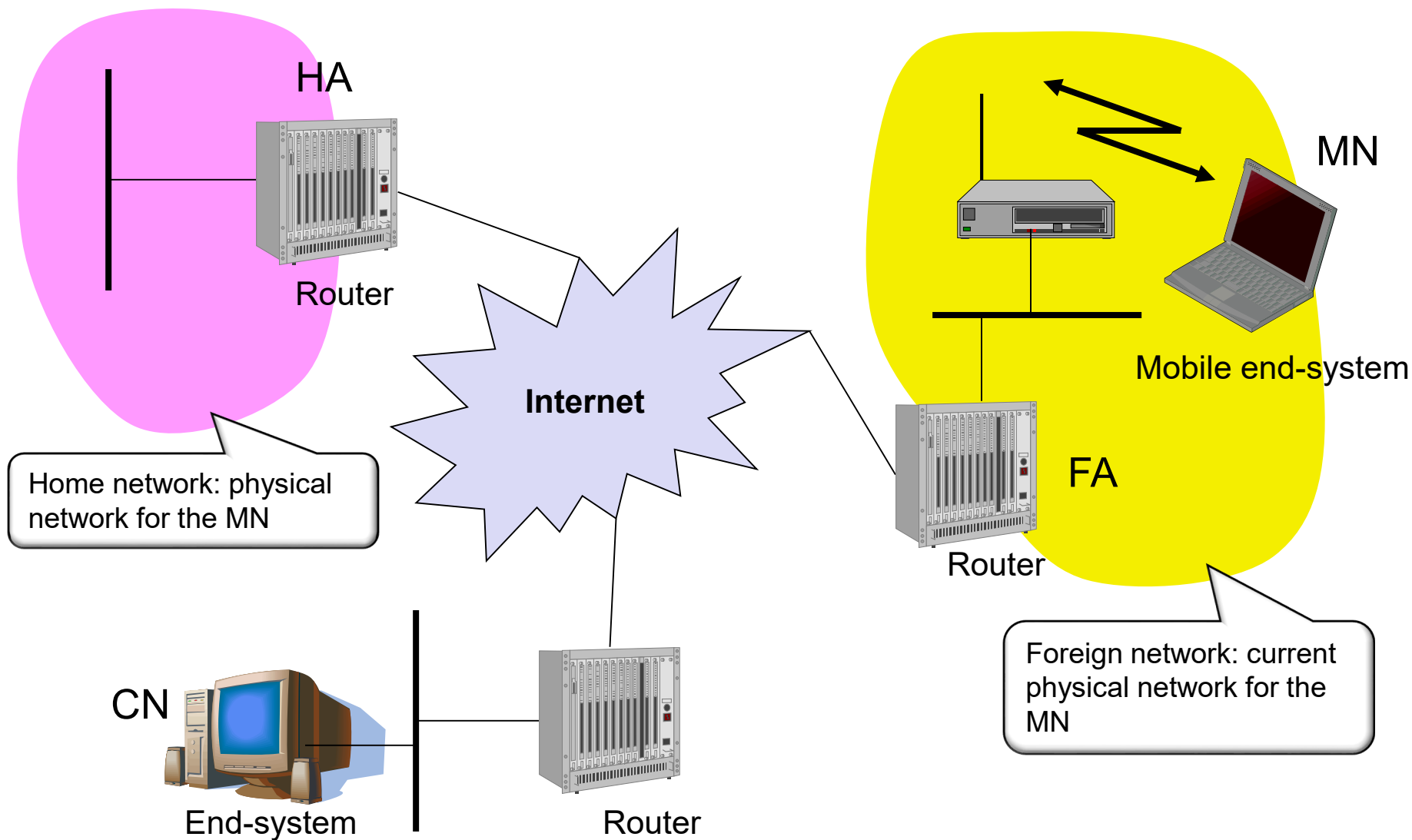
Mobile IP (Tunneling)

- Requirements for Mobile IP
 - Transparency
 - Mobile end-systems keep their IP address
 - Continuation of communication after interruption of link
 - Point of connection to the fixed network can be changed
 - Compatibility
 - Support of the same link layer protocols as IP
 - No changes to current end-systems and routers required
 - Mobile end-systems can communicate with fixed systems
 - Security
 - Authentication of all registration messages
 - Efficiency and scalability
 - Only little additional messages to the mobile system required (connection typically via a low bandwidth radio link)
 - World-wide support of a large number of mobile systems in the whole Internet

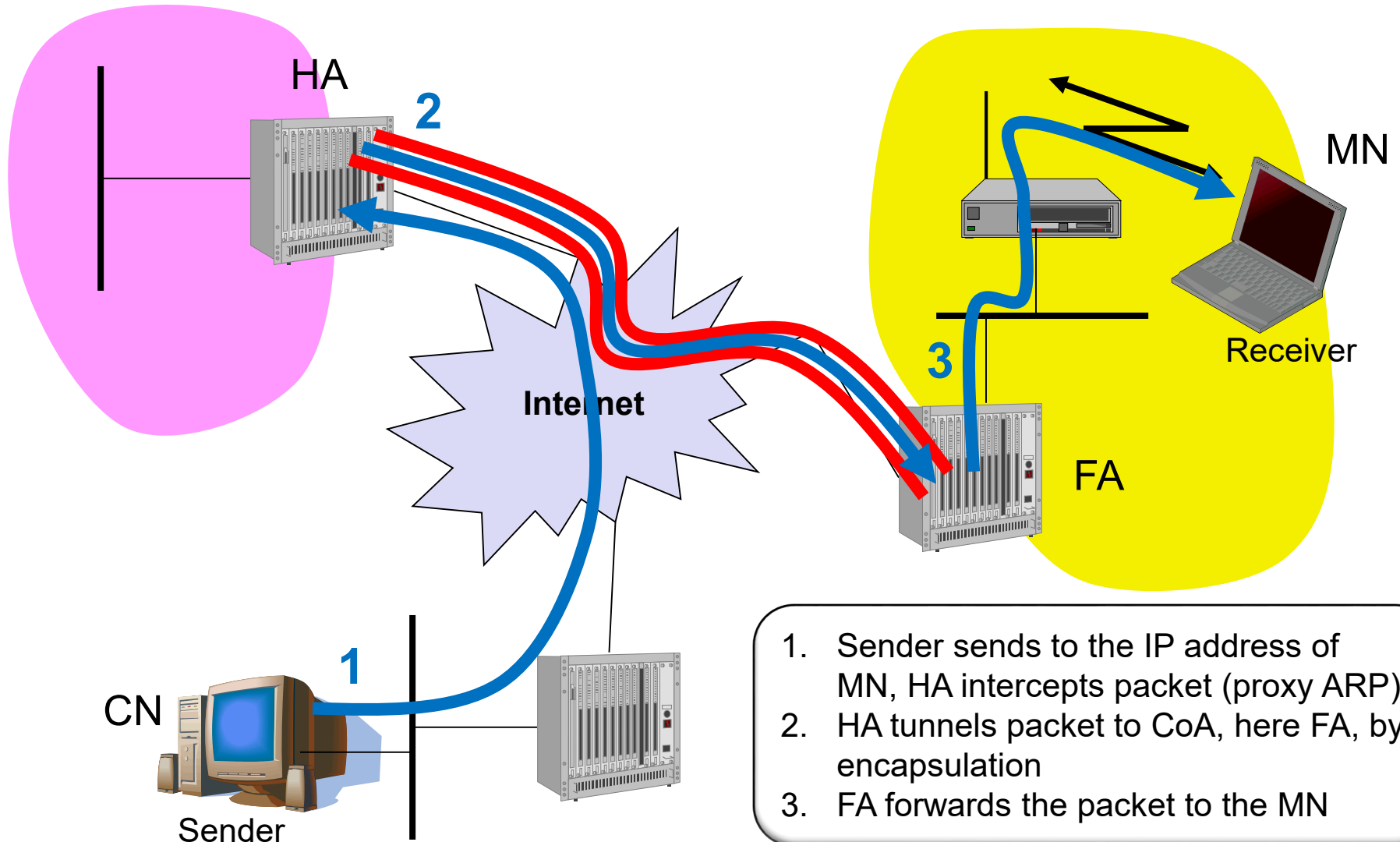
Mobile IP terminology

- **Mobile Node (MN)**
 - System (node) that can change the point of connection to the network without changing its IP address
- **Home Agent (HA)**
 - System in the home network of the MN, typically a router
 - Registers the location of the MN, tunnels IP datagrams to the CoA
- **Foreign Agent (FA)**
 - System in the current foreign network of the MN, typically a router
 - Forwards the tunneled datagrams to the MN, typically also the default router for the MN
- **Care-of Address (CoA)**
 - Address of the current tunnel end-point for the MN (at FA or MN)
 - Current location of the MN from an IP point of view
 - Can be chosen, e.g., via DHCP
- **Correspondent Node (CN)**
 - Communication partner

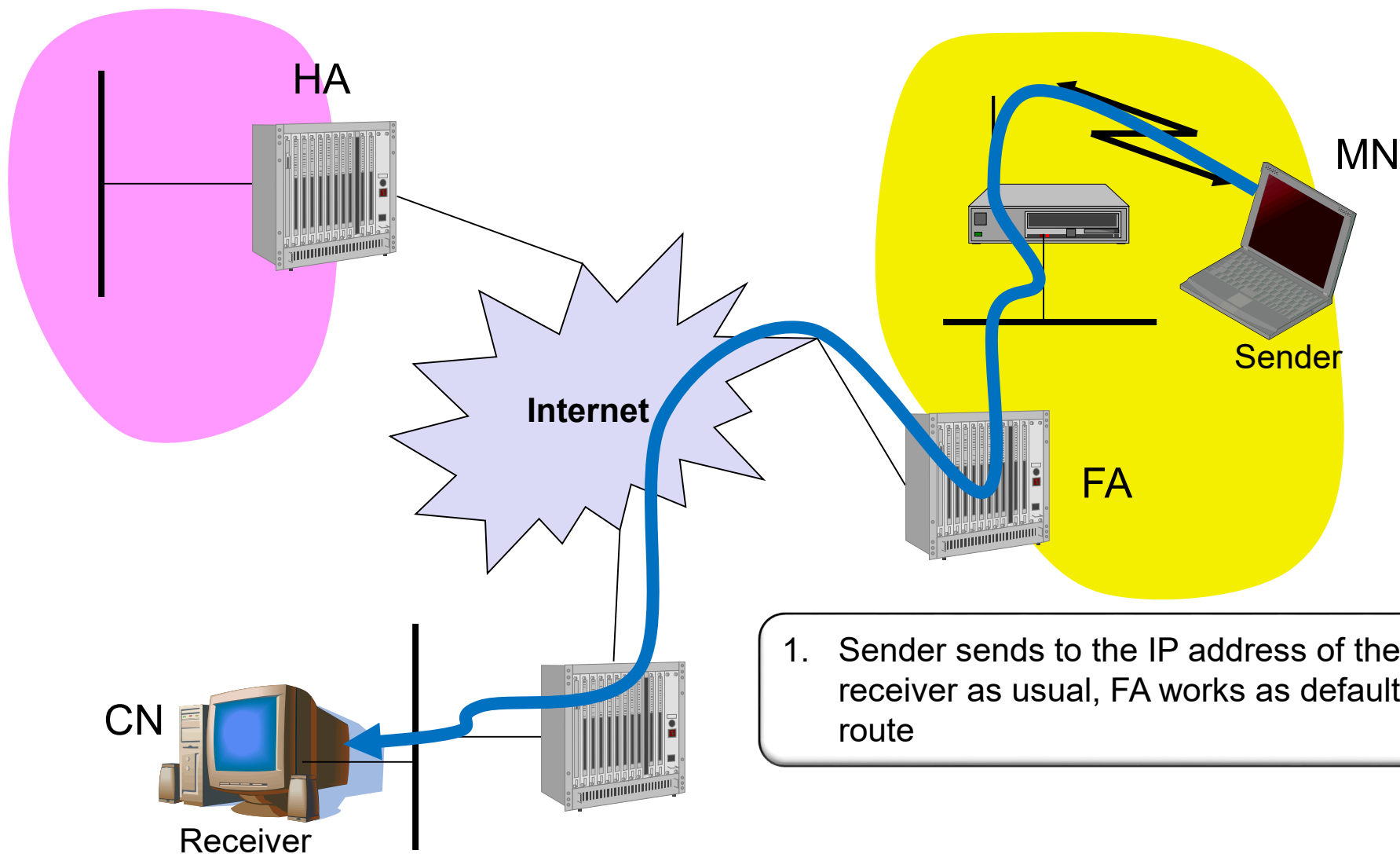
Mobile IP basics: example network



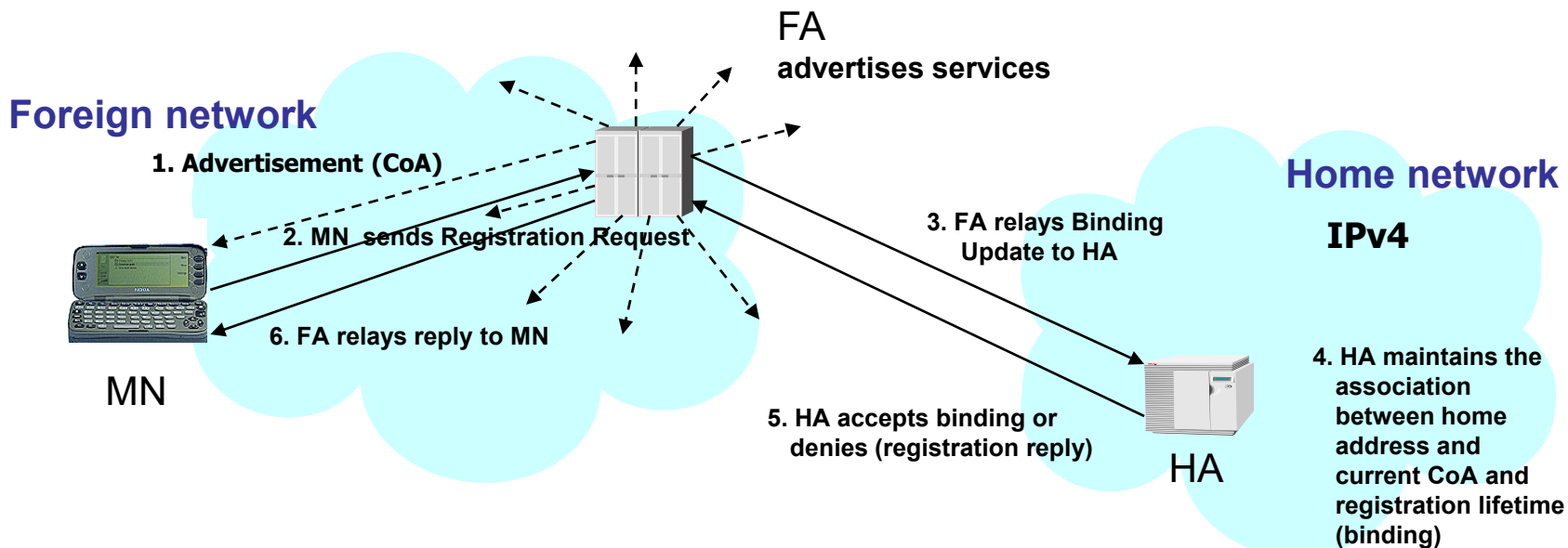
Mobile IP basics: data transfer to the mobile node



Mobile IP basics: data transfer from the mobile node



Mobile IP basics: agent discovery and registration (IPv4)

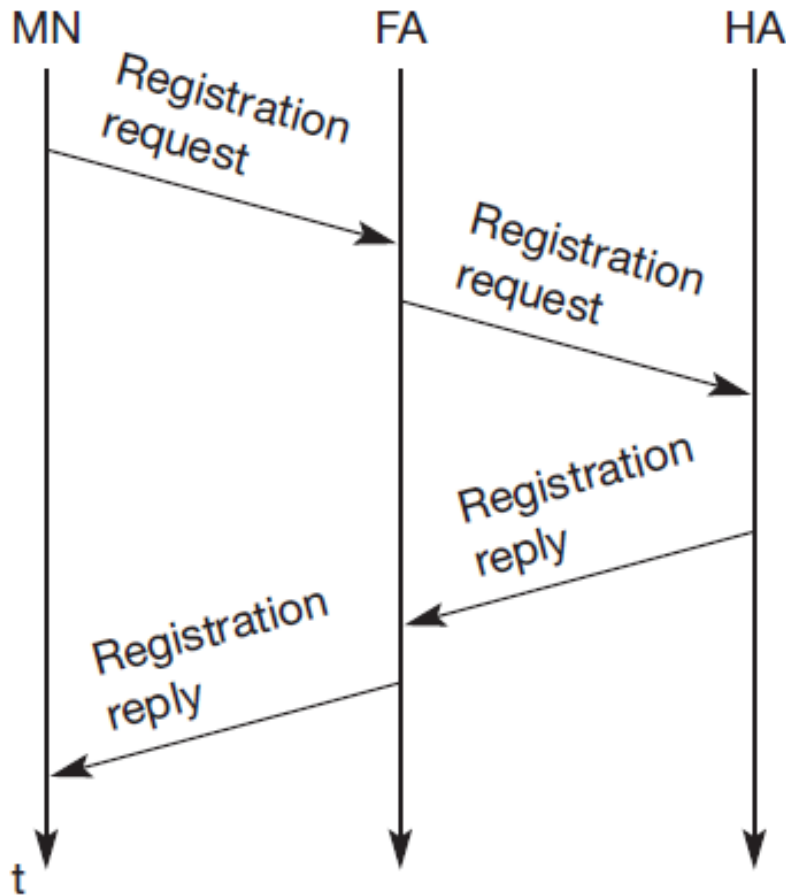


- Here the CoA denotes the address of the FA
 - The MN could also get it directly via DHCP ([co-located care-off address](#))
- The [binding update](#) is a remote redirect and therefore needs authentication
- After the registration, the HA tunnels data for the MN to the CoA

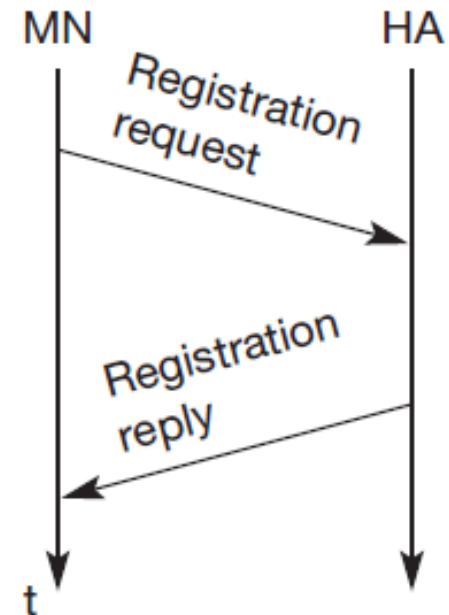
Mobile IP basics: agent discovery (IPv4)

- Agent discovery: How does the MN discover that it has moved?
 - Agent advertisement:
 - FAs and HAs advertise their presence periodically using special **agent advertisement** messages in their subnets
 - MN listens to these messages and detects, if it is in the home or a foreign network
 - MN reads a CoA from the FA advertisement messages
 - Agent solicitation:
 - If no agent advertisements are present or the inter-arrival time is too high, and an MN has not received a CoA by other means
 - The mobile node sends **agent solicitations** to prompt the transmission of agent advertisements

Mobile IP basics: registration (IPv4)



CoA is at the FA

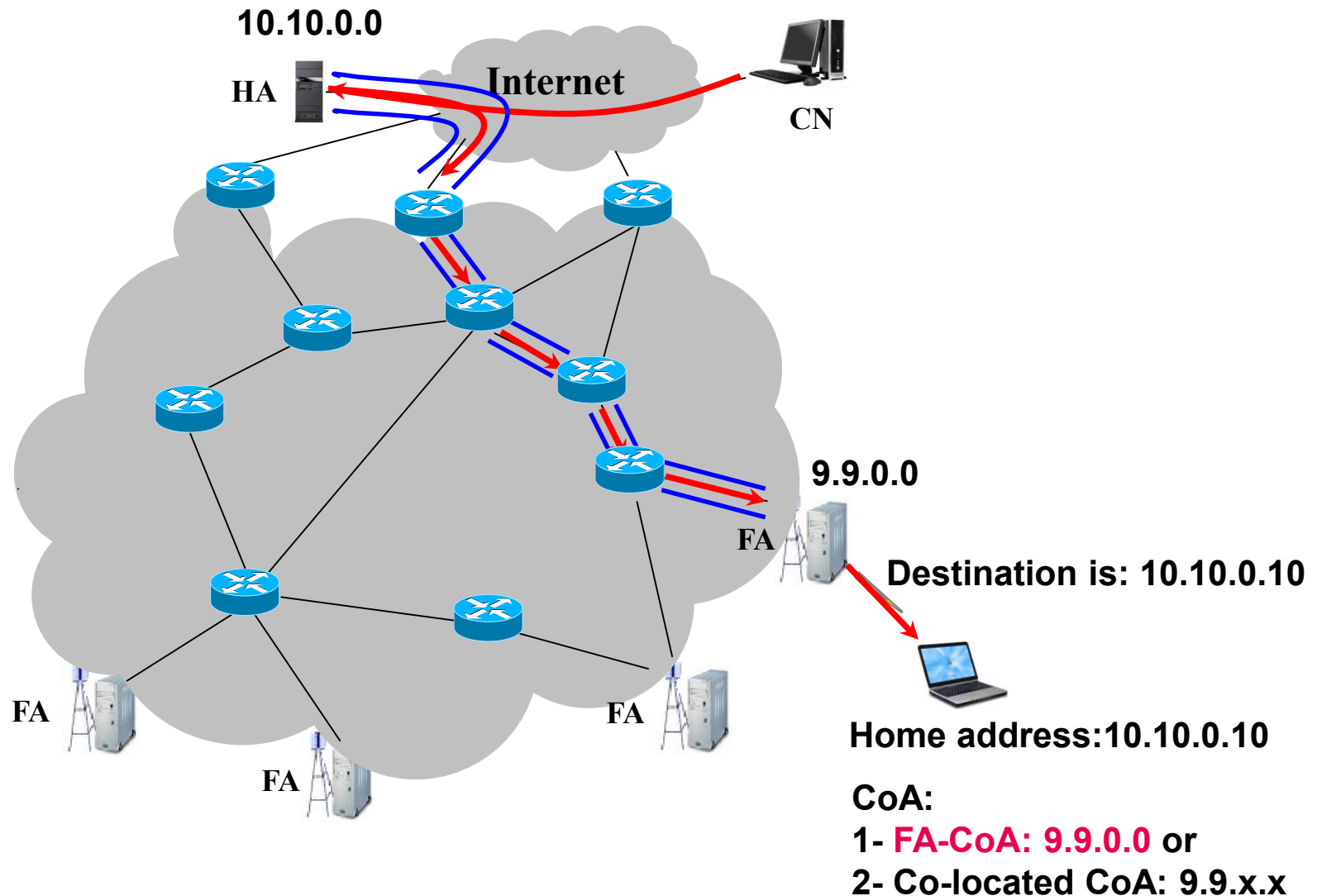


CoA is co-located

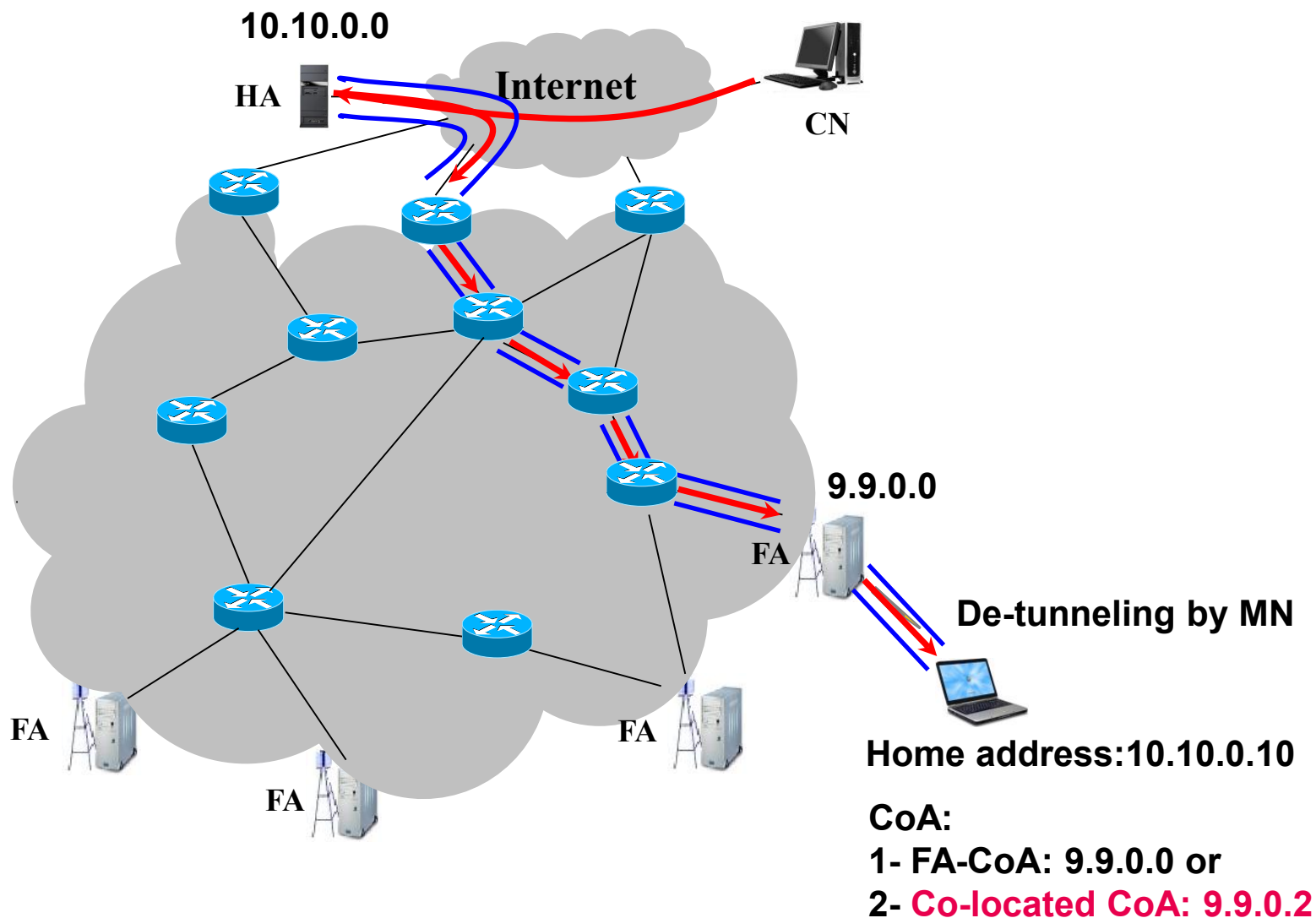
Mobile IP basics: registration (IPv4)

- Registration (always limited lifetime!)
 - Having received a CoA, the MN has to register with the HA
 - The main purpose of the registration is to inform the HA of the current location for correct forwarding of packets
- There are two ways of registration:
 - CoA is at the FA:
 - The MN sends its registration request containing the CoA to the FA
 - FA forwards the request to the HA
 - HA now sets up a **mobility binding** containing the mobile node's home IP address and the current CoA
 - After setting up the mobility binding, the HA sends a reply message back to the FA which forwards it to the MN
 - CoA is co-located with the MN:
 - MN sends the request directly to the HA and vice versa

Mobile IP basics: tunnel end points (IPv4)

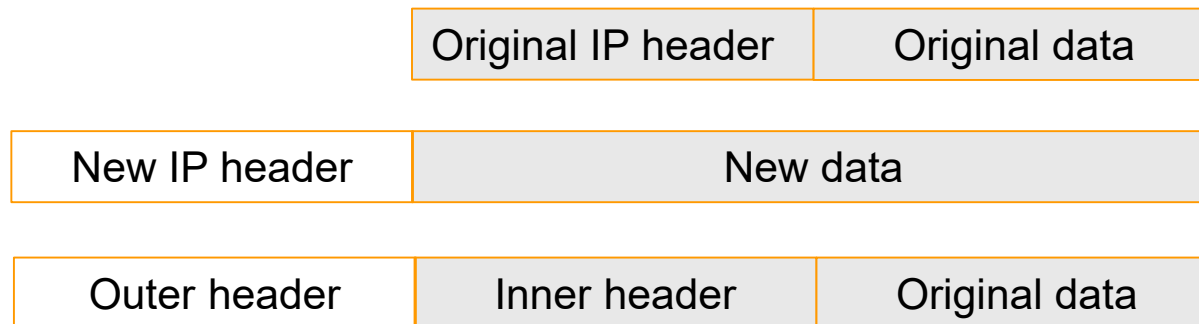


Mobile IP basics: tunnel end points (IPv4)



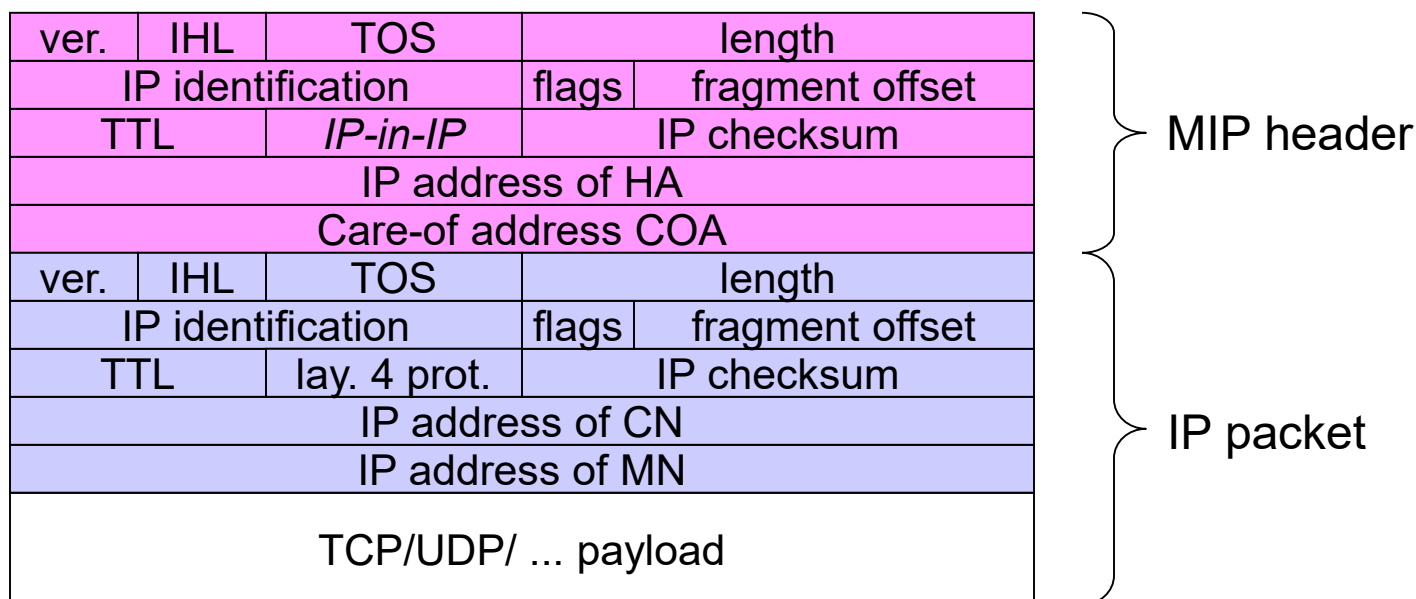
Tunneling – Idea

- Tunneling → sending a packet through a tunnel, is achieved by using encapsulation
- **Encapsulation** is the mechanism of taking a packet consisting of packet header and data and putting it into the data part of a new packet.
- **Decapsulation** is the reverse operation, taking a packet out of the data part of another packet.



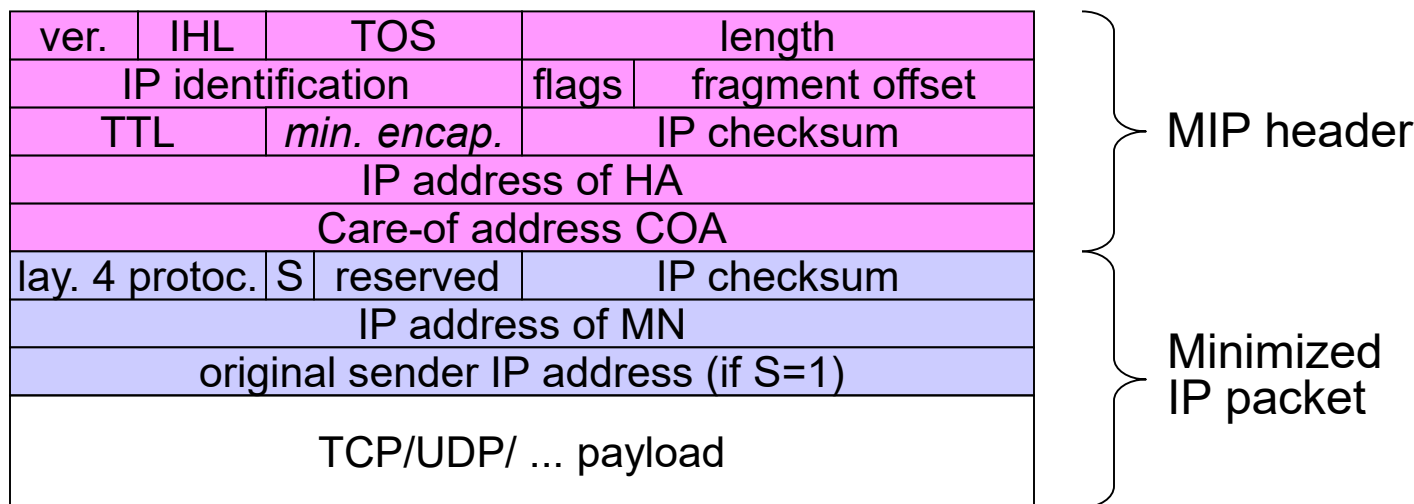
Tunneling – Details

- Encapsulation of one packet into another as payload
 - E.g. IPv6 in IPv4 (6Bone), Multicast in Unicast (Mbone)
 - Here: e.g. IP-in-IP-encapsulation, minimal encapsulation or GRE (Generic Record Encapsulation)
- IP-in-IP encapsulation (mandatory, RFC 2003)
 - Tunnel between HA and CoA



Tunneling – Details

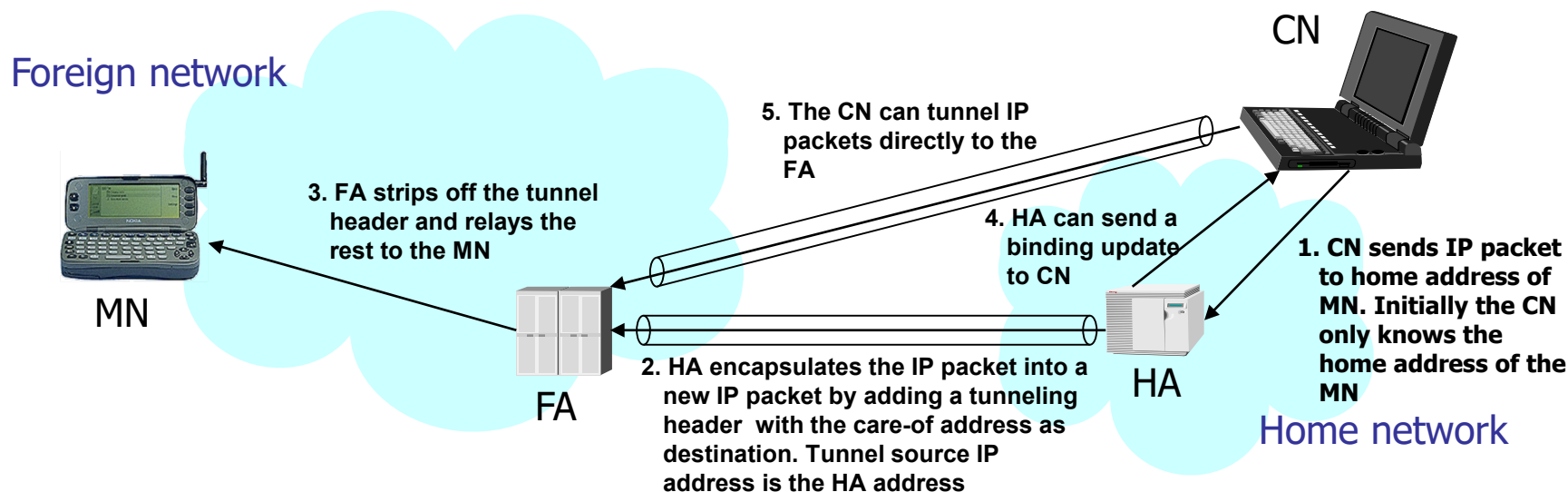
- Minimal encapsulation (optional)
 - Avoids repetition of identical fields
E.g. TTL, IHL, version, TOS
 - Only applicable for unfragmented packets, no space left for fragment identification



Optimization of packet forwarding

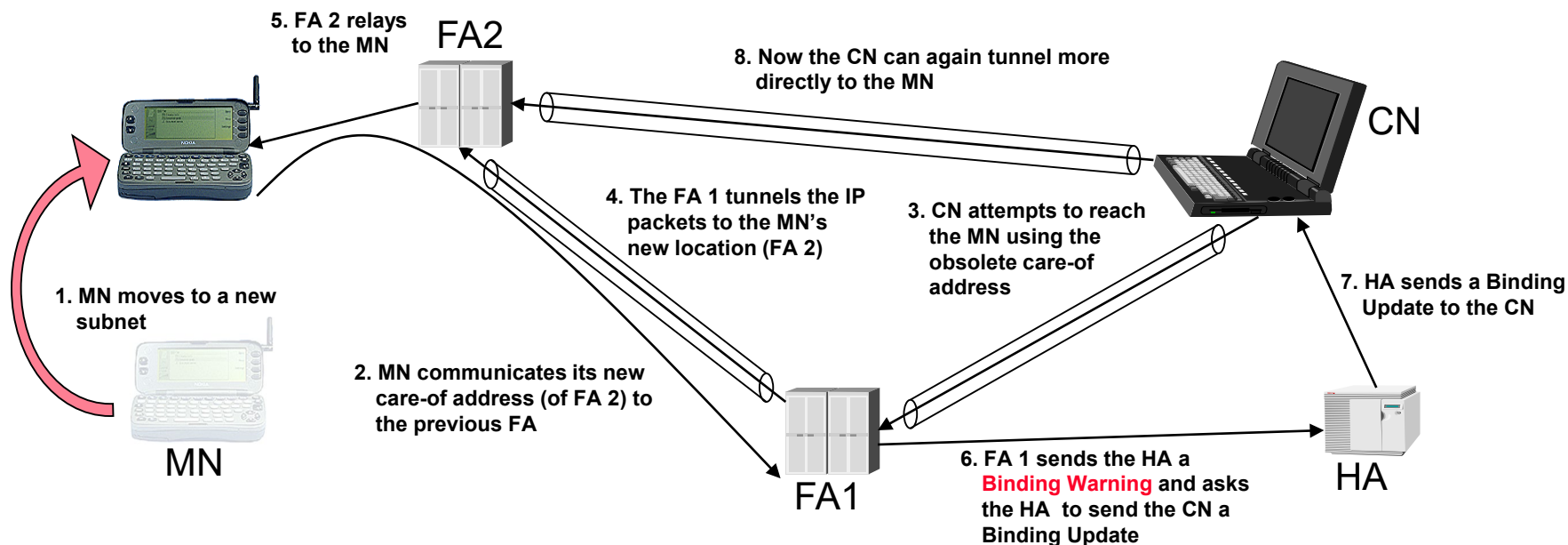
- Triangular Routing
 - CN sends all packets via HA to MN
 - Higher latency and network load
- Solution Ideas
 - CN learns the current location of MN
 - Direct tunneling to this location
 - HA informs a sender about the location of MN
 - Security problems with binding updates
- Change of FA
 - Packets on-the-fly during the change can be lost
 - New FA informs old FA to avoid packet loss, old FA now forwards remaining packets to new FA
 - This information also enables the old FA to release resources for the MN

Tunneling and route optimization (IPv4)



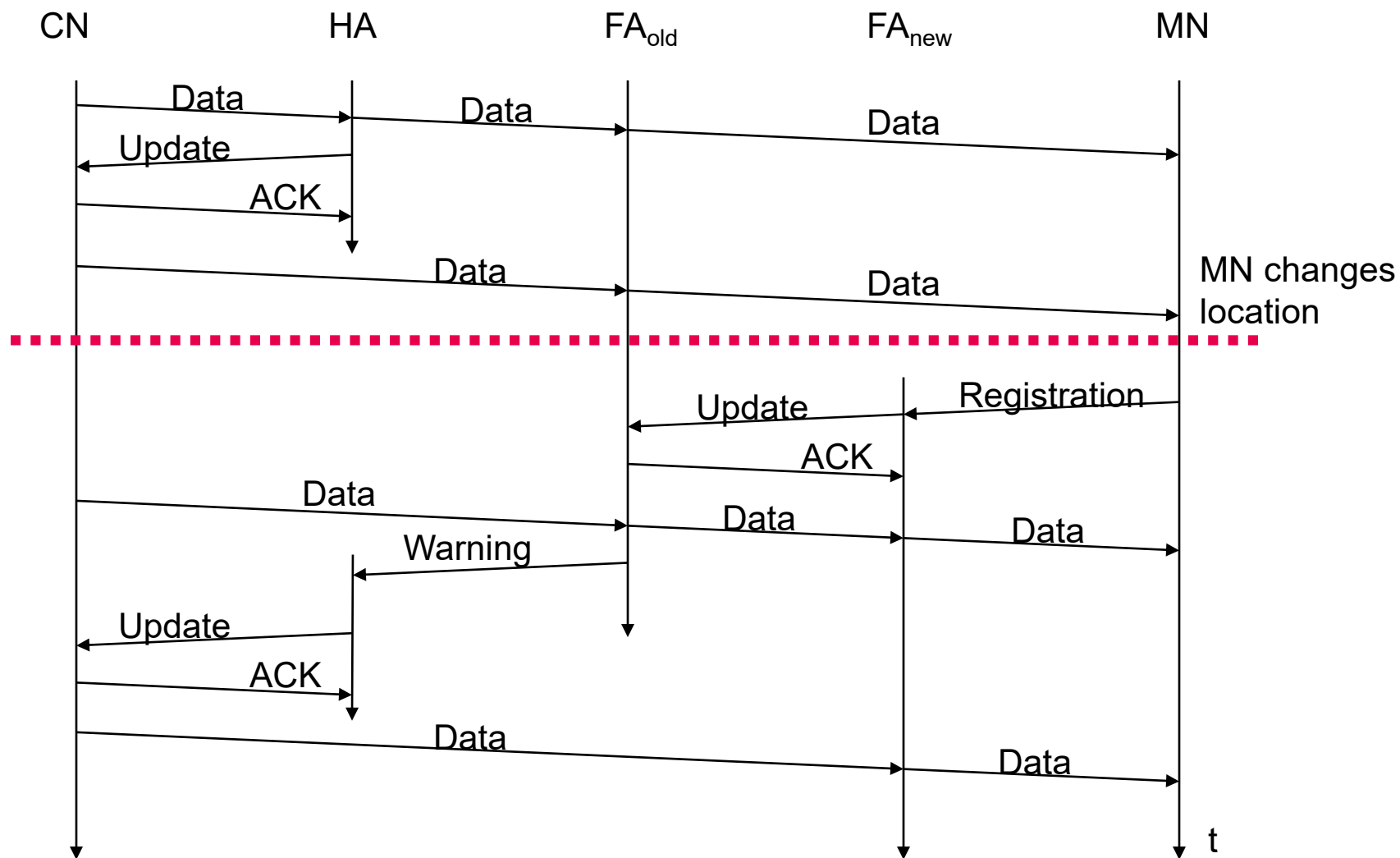
- The home agent can either use IP-within-IP or minimal encapsulation to reduce the header overhead
- **Route optimization** can be done if the HA sends a Binding Update to the CN
 - Subsequent packets can be tunneled directly from the CN to the FA
- In the reverse direction normal standard IP routing mechanisms are used to deliver IP datagrams from the MN
- Problems if the MN moves to another FA!

Change of foreign agent (IPv4)

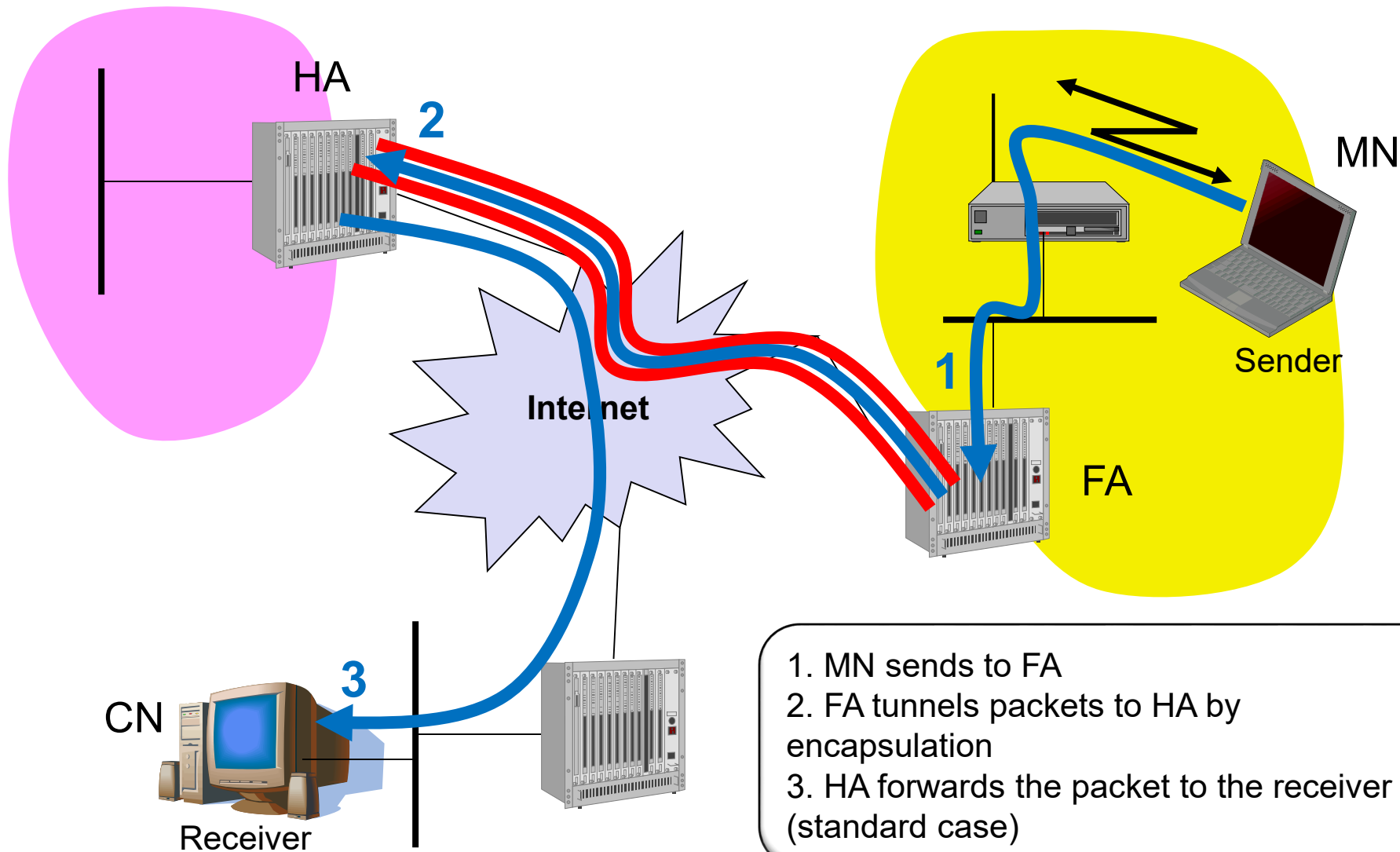


- After having moved to the new FA, the MN **can** also communicate its new care-of address to the previous FA
- Note that most of the Binding Update messages are not emitted by the MN, thus they are not going over the wireless link
- Problem if FA1 does not know the new care-of address of the MN
 - FA1 must re-route the traffic to the HA, which again handles it via a new tunnel to FA2

Change of foreign agent (IPv4)



Reverse tunneling (RFC 3024)



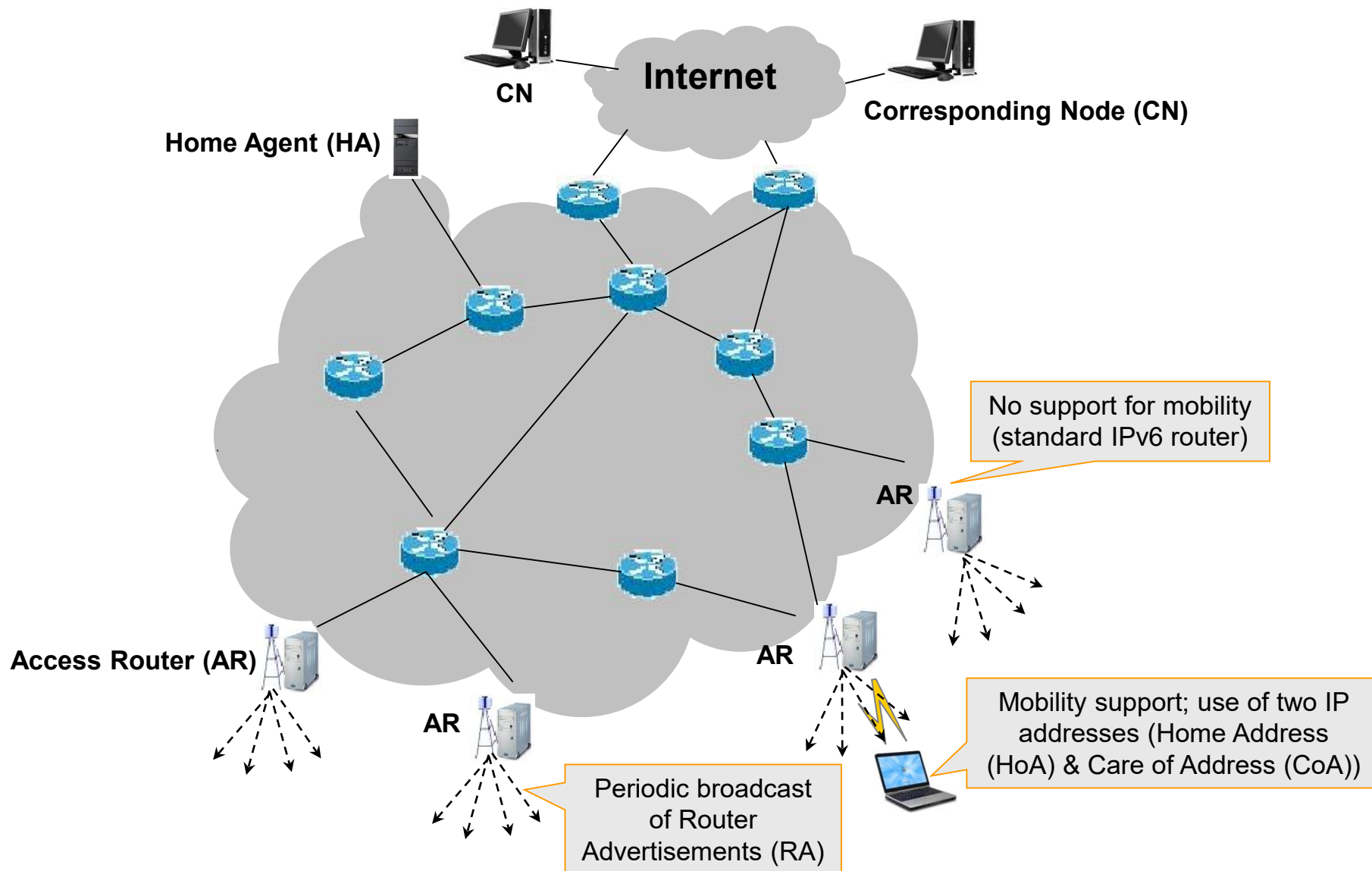
Mobile IP with reverse tunneling

- Router accept often only “topological correct” addresses (firewall!)
 - A packet from the MN encapsulated by the FA is now topological correct
 - Furthermore multicast and TTL problems solved (TTL in the home network correct, but MN is too far away from the receiver)
- Reverse tunneling does not solve
 - Problems with *firewalls*, i.e. the reverse tunnel can be abused to circumvent security mechanisms (tunnel hijacking)
 - Optimization of data paths, i.e. packets will be forwarded through the tunnel via the HA to a sender (double triangular routing)
- Mobile IP reverse tunneling is backwards compatible
 - The extensions can be implemented easily and cooperate with current implementations of Mobile IP without these extensions
 - Agent advertisements can carry requests for reverse tunneling

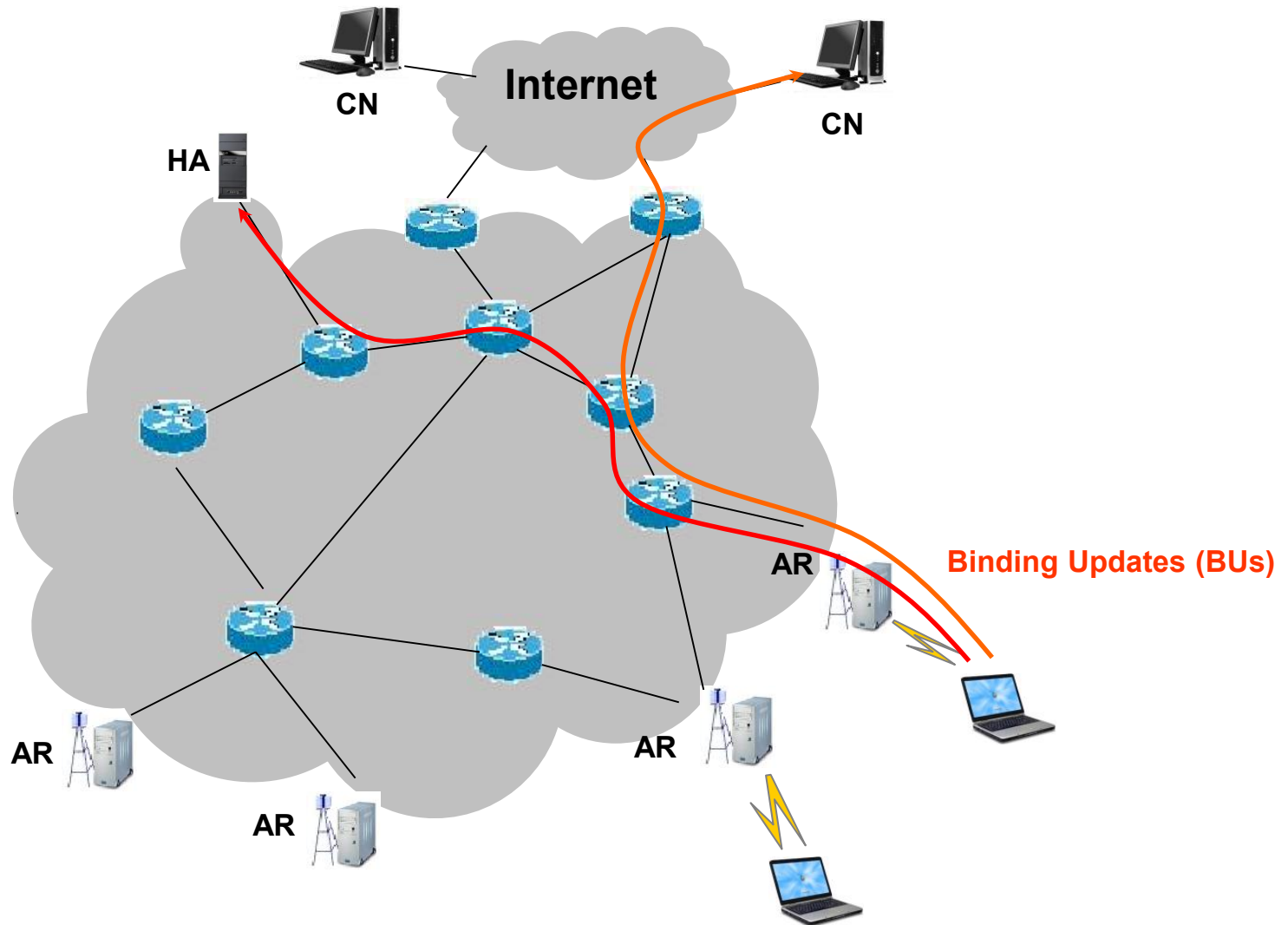
Mobile IP and IPv6

- Huge address space with IPv6 address renumbering scheme
- Better mobility support
 - Mobile IP was designed as add-on to IPv4
 - Majority of IPv4 nodes do not support Mobile IP
- MN can use **Address Autoconfiguration** which eliminates the need of a Foreign Agent
 - **Stateless**: to configure its care-of address from a NW prefix of the foreign NW and a MN interface identifier. The needed information is published by neighboring routers through the **Neighbor Discovery** protocol in the foreign network (combined ARP/ICMP)
 - **Stateful**: by using a centralized DHCP server
- **Source Routing (source defines the route)**: Instead of tunneling packets using “IP in IP”, IPv6 Routing Headers are used
- IPv6 does work smoothly with **ingress filtering**
 - In IPv4 a border router may discard IP packets not originating from its own subnet, and MN use their home address always as source address
- Every IPv6 node will have implemented IPv6 **authentication headers** to be used with binding updates

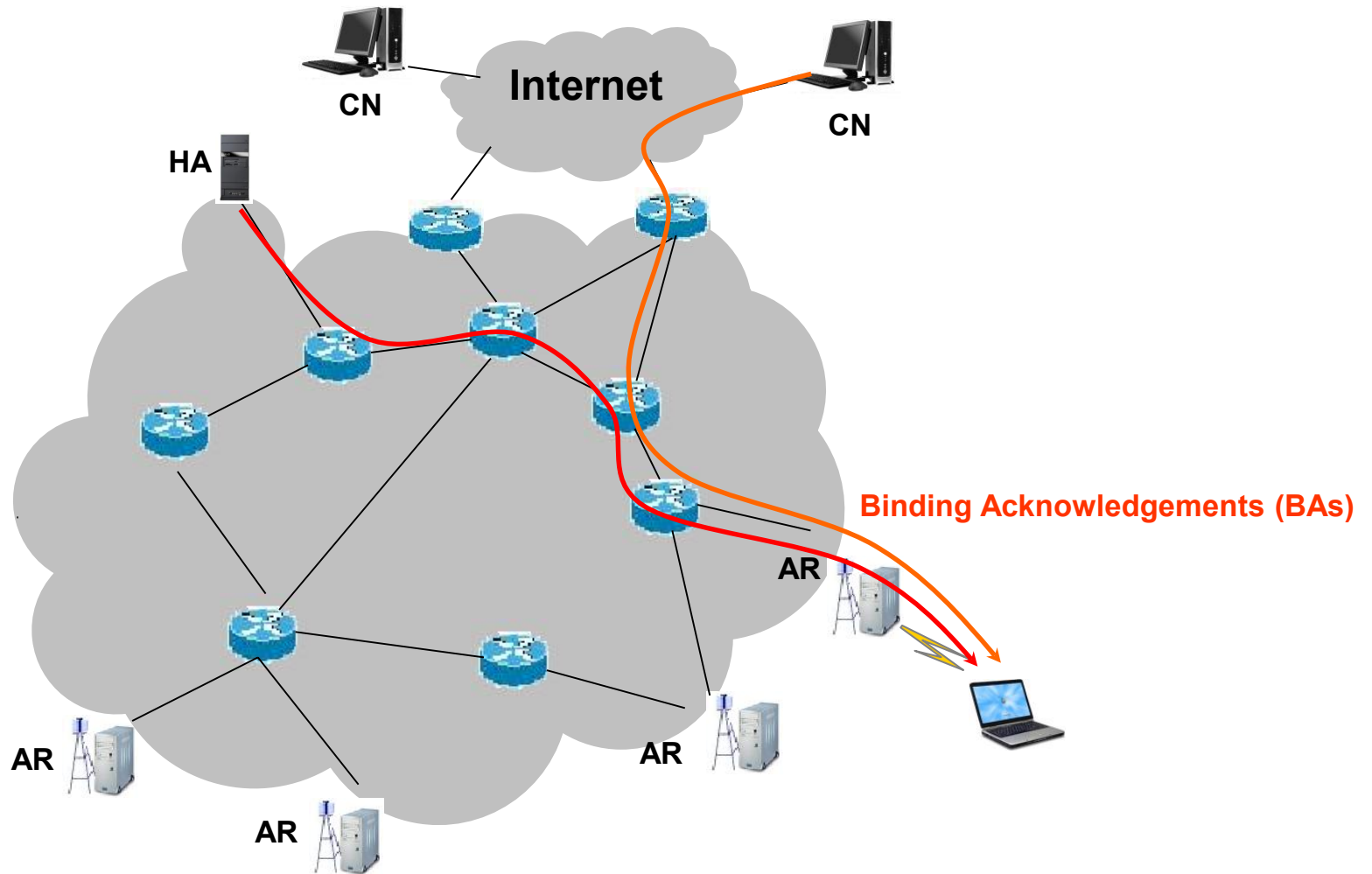
Mobile IPv6 (MIPv6)



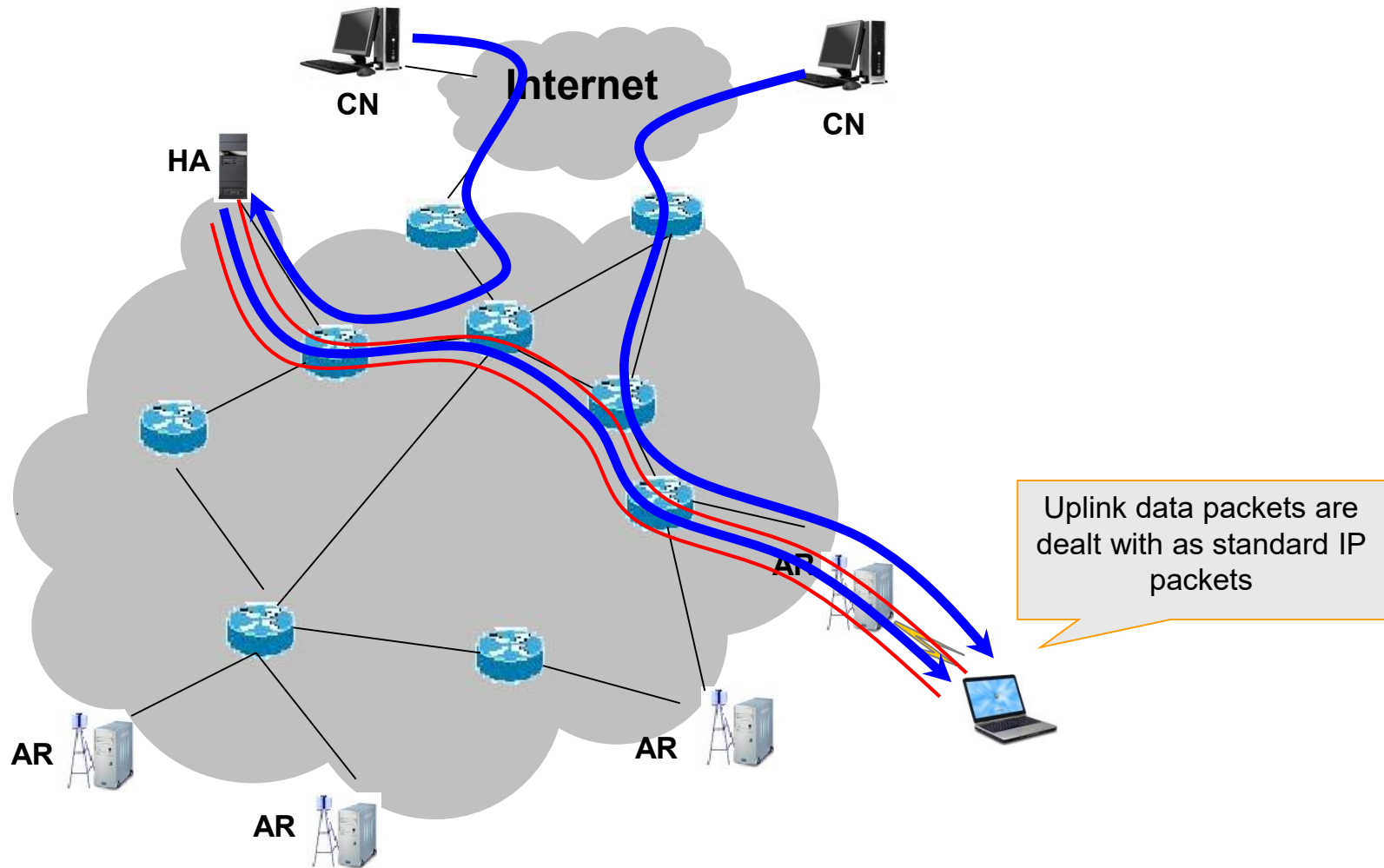
MIPv6 - Handoff



MIPv6 - Handoff



MIPv6 – Data Communication



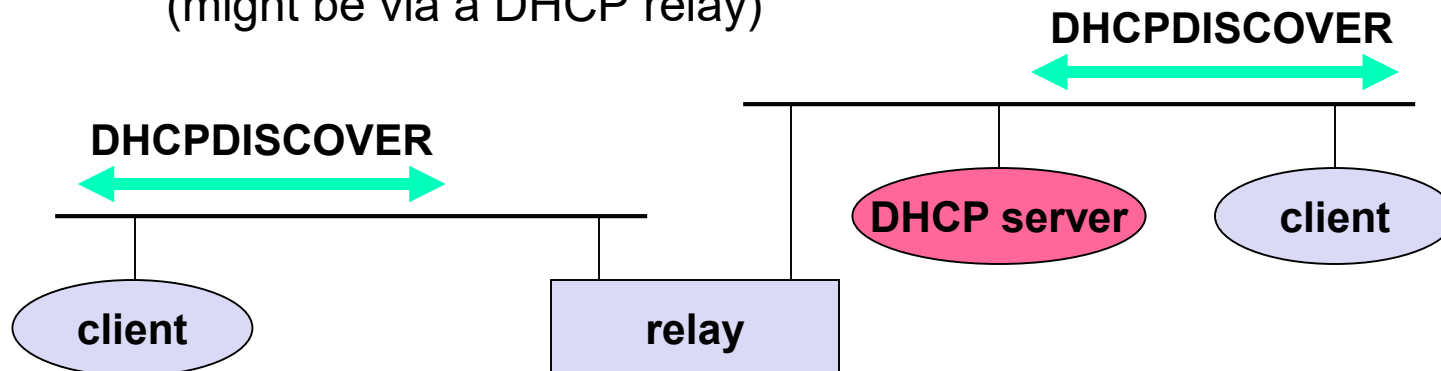
MIPv6 – Pros & Cons

- Pros
 - End-to-end mobility
 - Transparency to upper layer protocols
 - Proper integration with IP

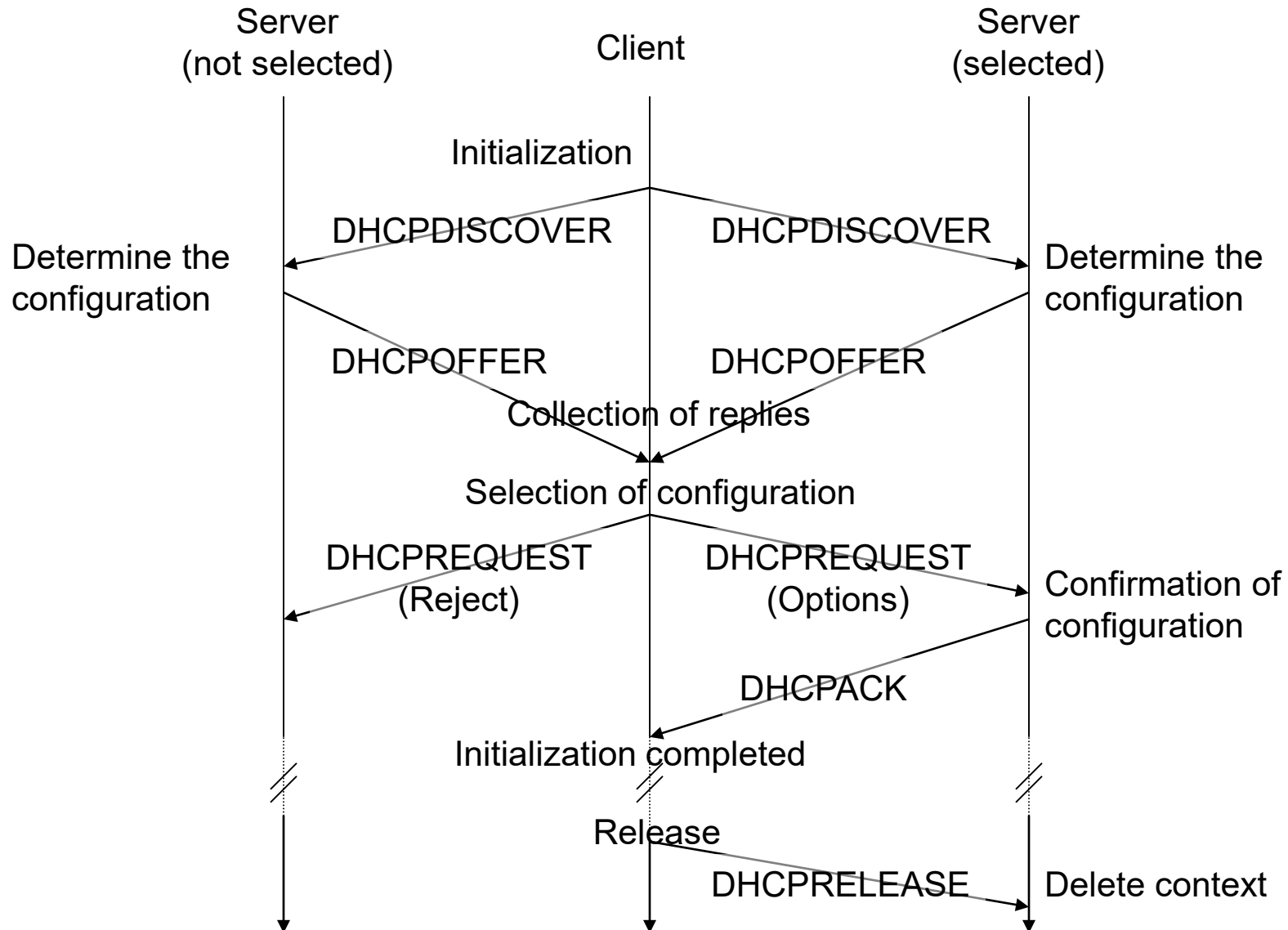
- Cons
 - Handoff latencies (traffic interruption) due to contacting the HA and the CN each time the MN changes the point of attachment
 - Overhead due to encapsulation
 - Signaling load due to frequent BUs especially when MNs move at high speeds
 - Sub-optimal routing and increased end-to-end delay due to triangular routing where CN is not MIPv6 enabled (binding updates) or due to lack of security

Dynamic Host Configuration Protocol (DHCP)

- Application
 - Simplification of installation and maintenance of networked computers
 - Supplies systems with all necessary information, such as IP address, DNS server address, domain name, subnet mask, default router etc.
 - Enables automatic integration of systems into an Intranet or the Internet
 - Can be used to acquire a CoA for Mobile IP (co-location of MN and FA)
- Client/Server-Model
 - The client sends via a MAC broadcast a request to the DHCP server (might be via a DHCP relay)



DHCP – protocol mechanisms



DHCP characteristics

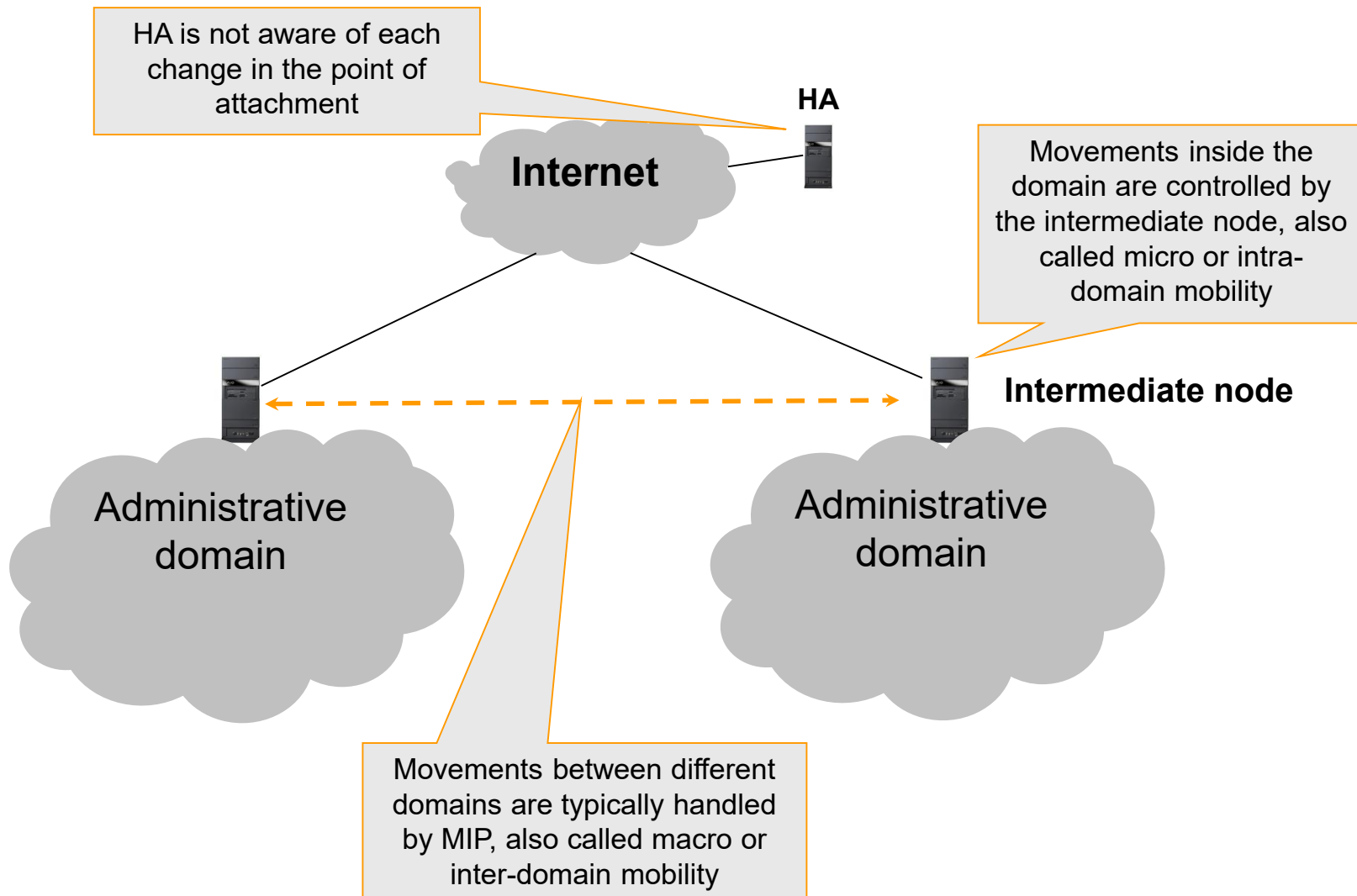
- Server
 - Several servers can be configured for DHCP, coordination not yet standardized (i.e. manual configuration)
- Renewal of configurations
 - IP addresses have to be requested periodically, simplified protocol
- Options (RFC 2132)
 - Available for routers, subnet mask, NTP (network time protocol) timeserver, SLP (service location protocol) directory, DNS (domain name system)
- Big security problems!
 - No authentication of DHCP information specified

Network Layer MM – Advanced solutions

Questions to be answered:

- Identify the limitations of Mobile IP. Which optimizations can you think of?
- What is the purpose of these optimizations?
- Which problem is addressed by Hierarchical MIP?
- What is the idea of HMIP?
- Which problem is addressed by Cellular IP?
- What is the idea of Cellular IP?
- Why do updates of the routing table make sense here?
- Can Cellular IP be generalized to non-hierarchical systems?
- Which problem is addressed by Proxy MIP?
- What is the idea of Proxy MIP?

Macro and Micro Mobility Management



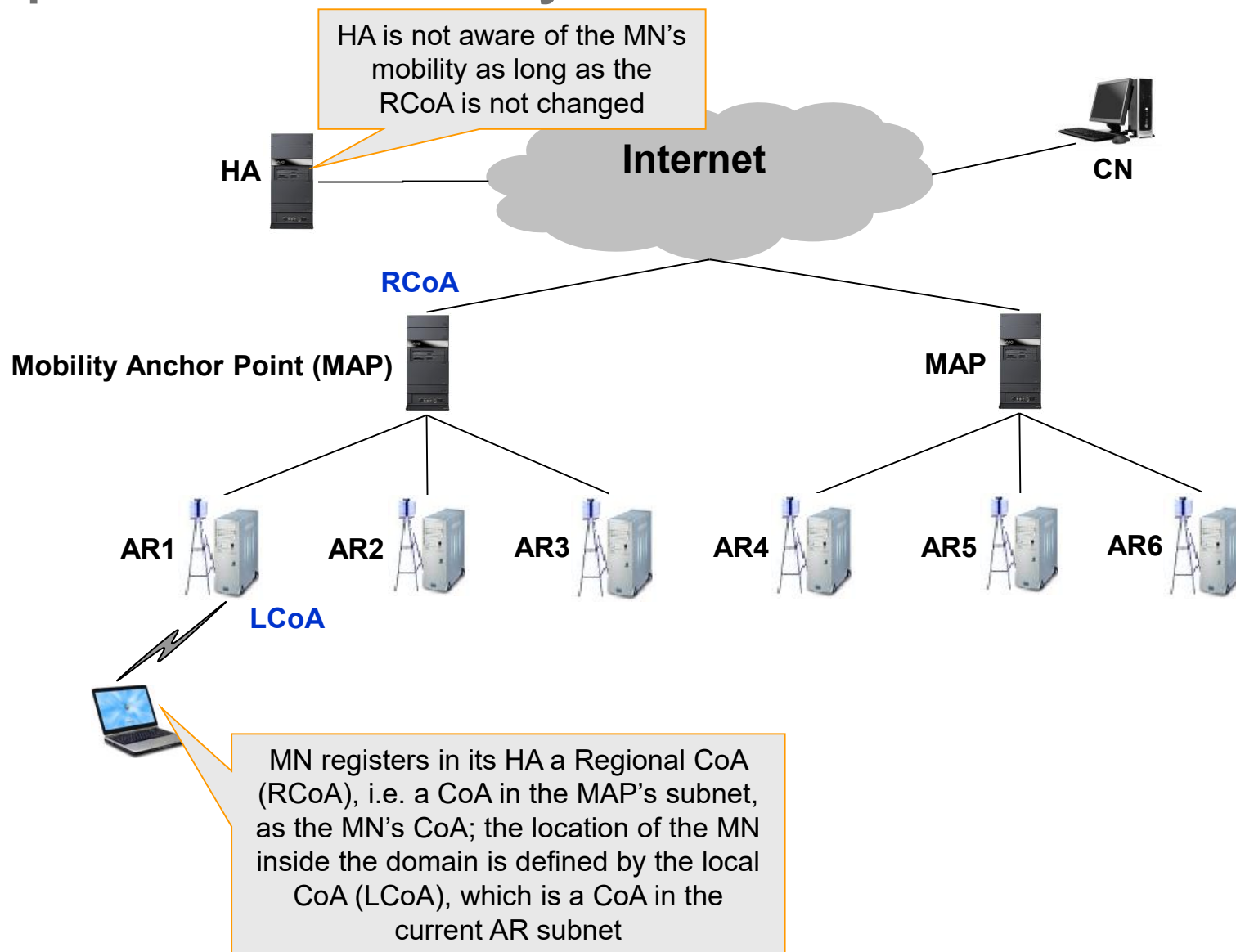
Macro and Micro Mobility Management

- Macro mobility management approaches
 - Aim at global mobility support
 - Base protocols: Mobile IPv4 (RFC 3344) & Mobile IPv6 (RFC3775)
- Micro mobility management approaches
 - Performance improvements through localizing the mobility processing inside administrative domains
 - Introduction of new intermediate nodes to the network
 - Hierarchical network topology is typically required

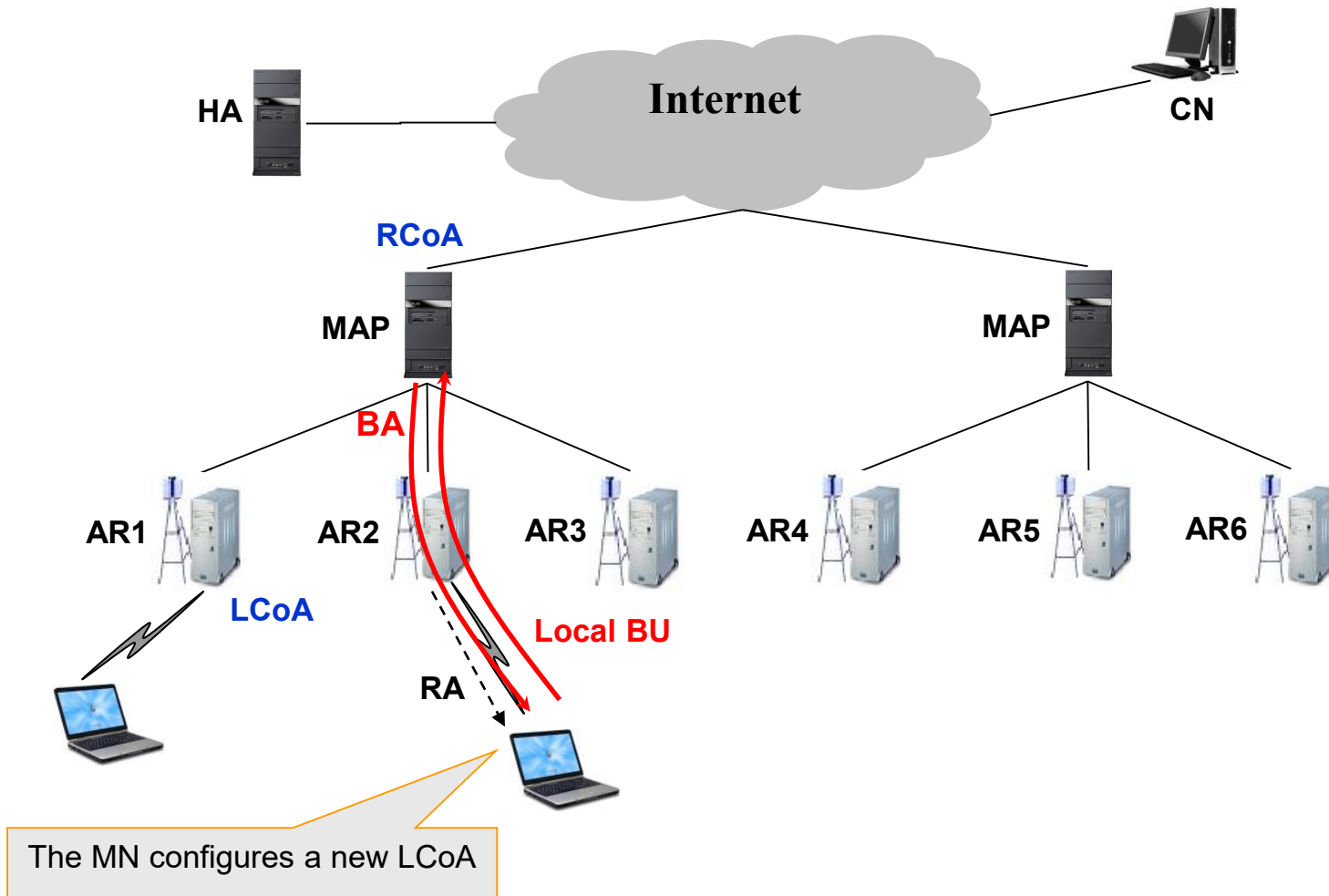
Proxy Agent Approaches (PAA)

- Extend the principle of MIP to provide micro mobility
- Introduce new intermediate nodes (regional agents) to the network to process the mobility inside an administrative domain locally
- Data packets destined to the MN are forwarded to the intermediate node, which **tunnels them towards the MN's current point of attachment**
- Examples
 - Hierarchical MIPv6 (HMIPv6), Anchor Foreign Agent (AFA), Seamless MIP (S-MIP), etc.

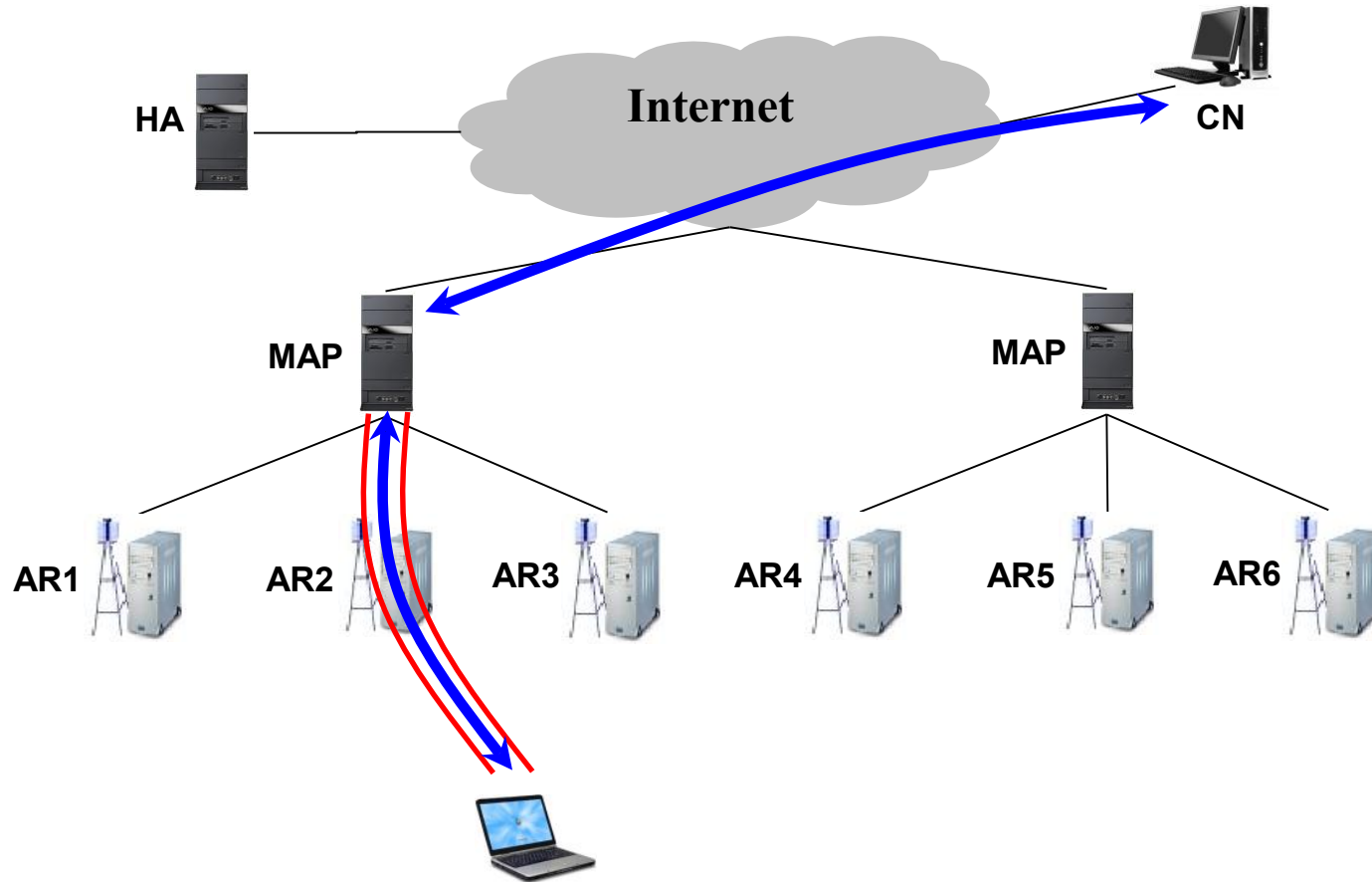
Improved Micro Mobility: Hierarchical MIPv6



HMIPv6 – Handoff



HMIPv6 – Data Communication



HMIPv6 – Pros & Cons

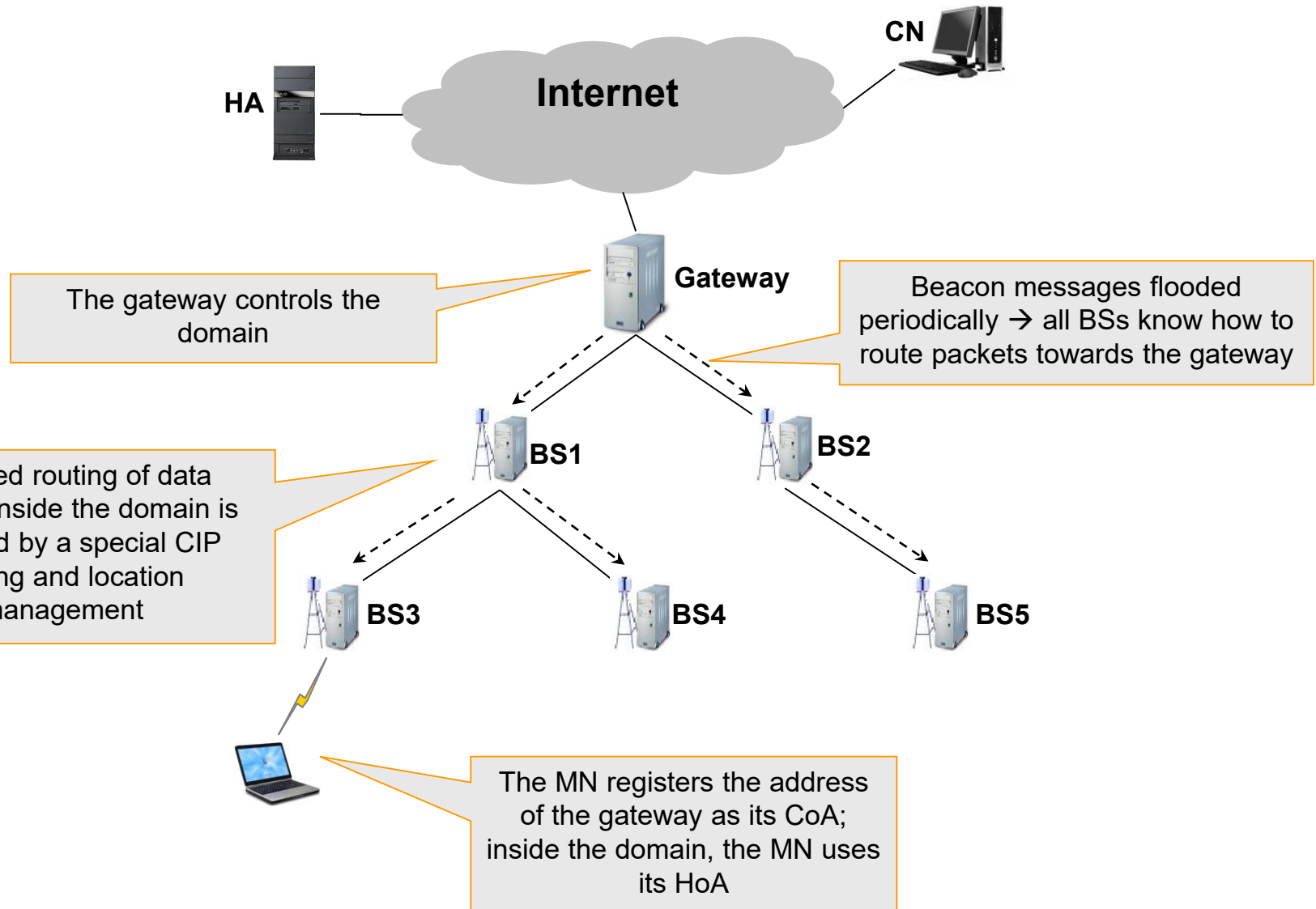
- Pros
 - Localizing of mobility management inside administrative domains
 - Improving the performance: reduced handoff latency compared to MIPv6, reduced packet loss, etc.
 - Reducing the signaling load traveling towards the HA

- Cons
 - Restrictions on the network topology
 - Single point of failure (all traffic passes through the MAP)
 - Increased signaling and packet forwarding overhead inside the domain
 - Security issues: how to authenticate arriving MNs?

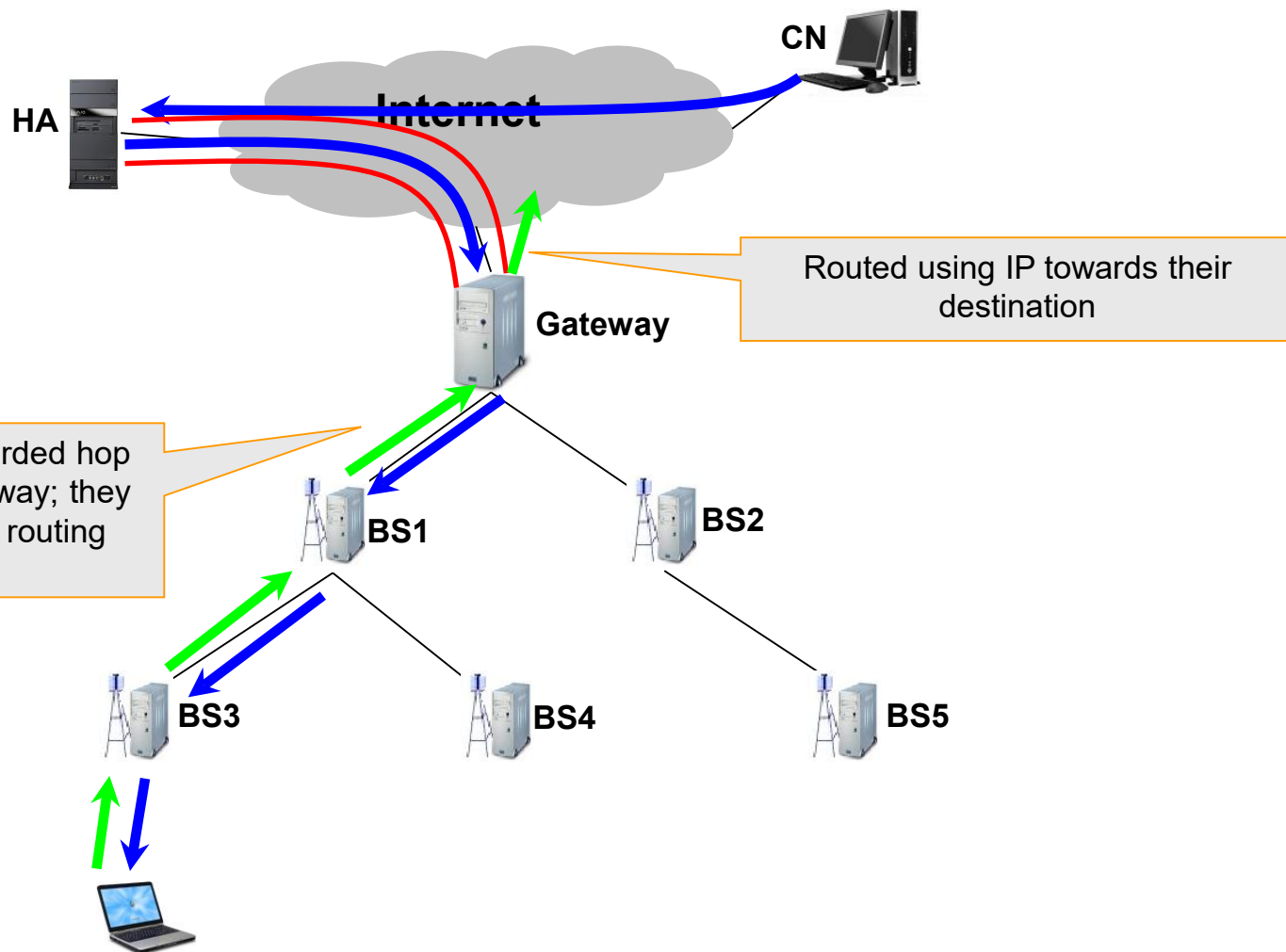
Localized Enhanced Routing Schemes

- Similar principle as PAA approaches
- Typically introduce a new dynamic layer 3 routing protocol inside the domain (per-hop routing)
- Data packets destined for the MN are forwarded to the intermediate node controlling the domain, which **forwards them hop per hop towards the MN**
- All nodes in the domain should be mobility-aware
- Examples
 - Cellular IP, HAWAII

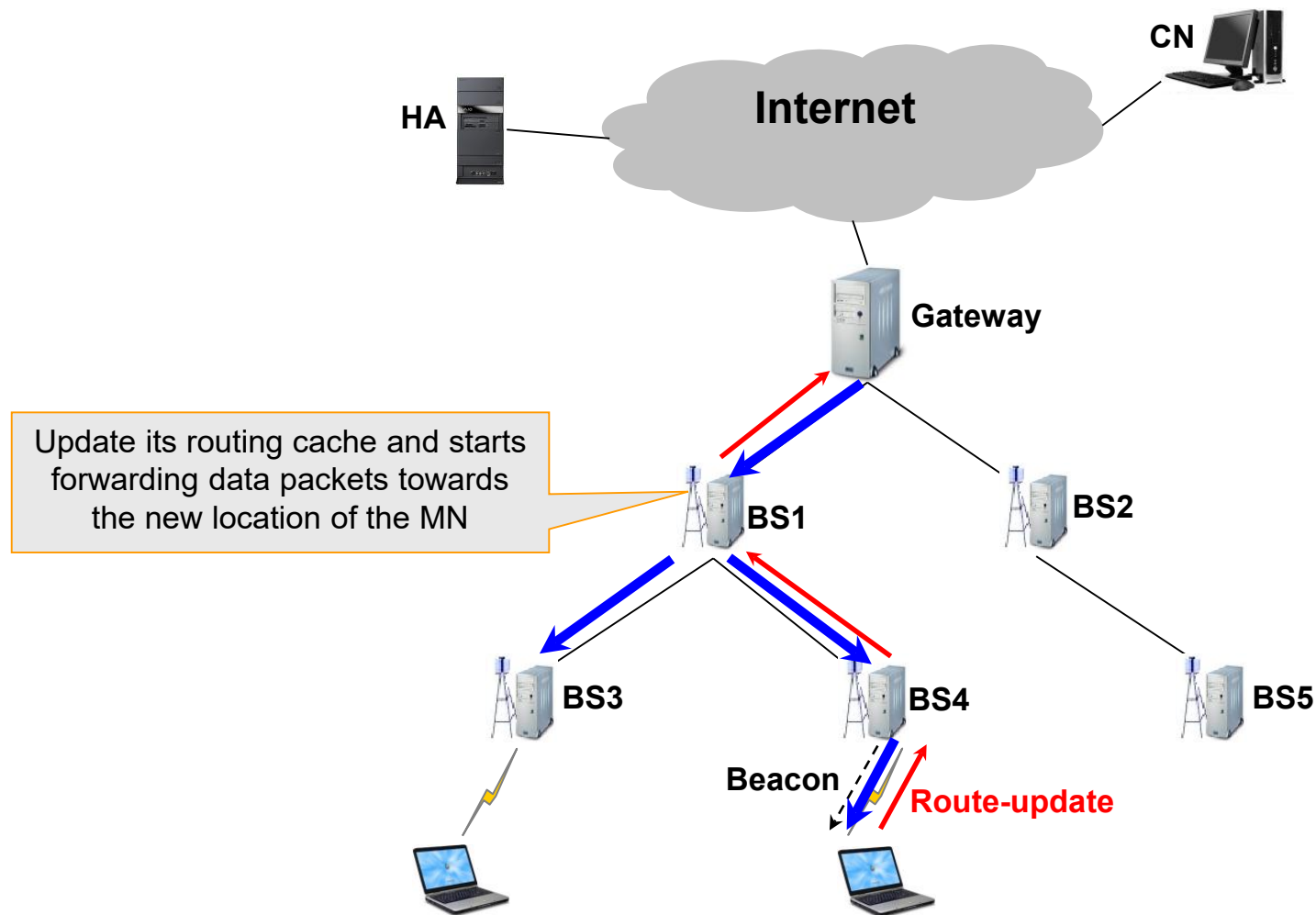
Improved Micro Mobility: Cellular IP



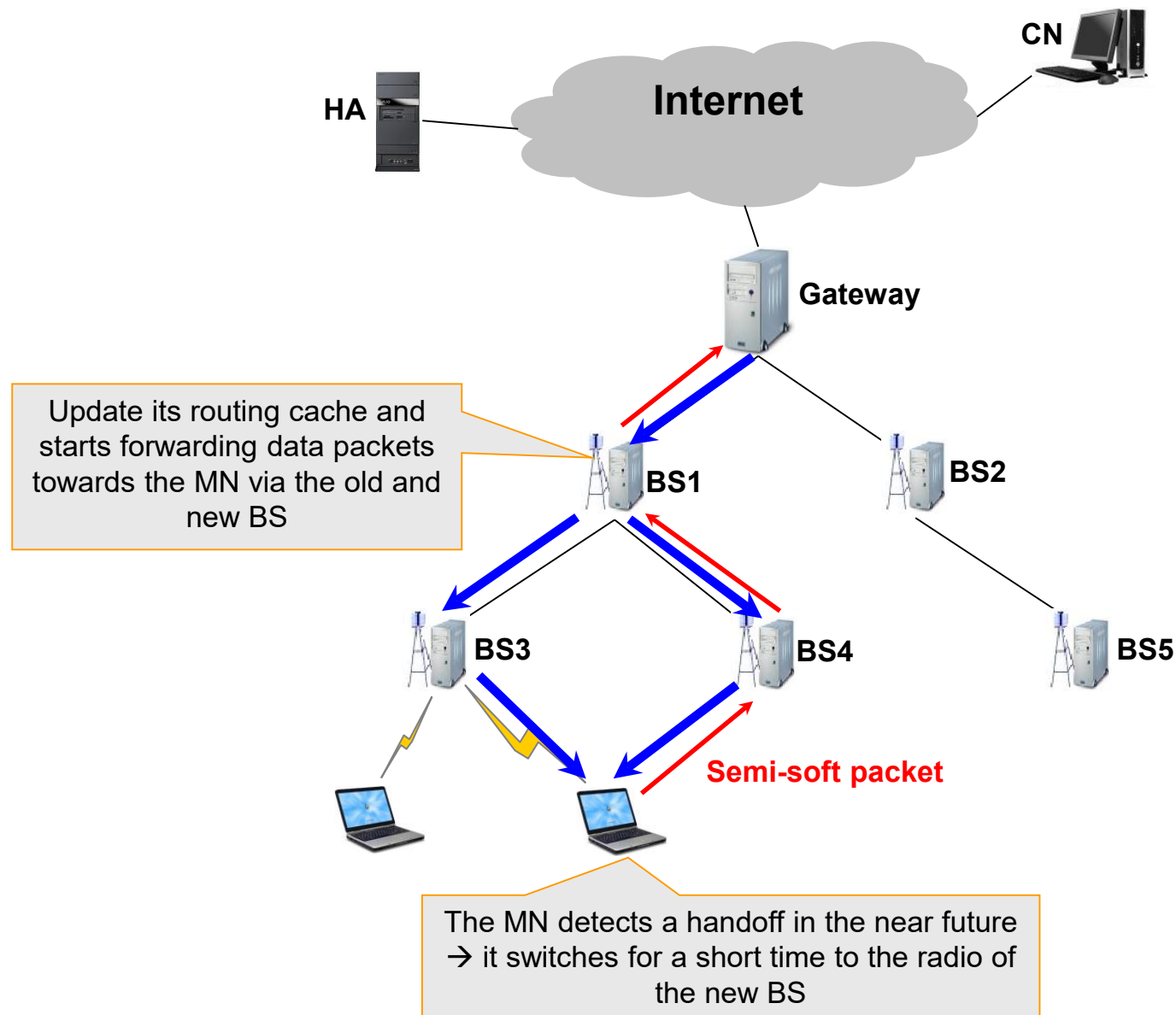
Cellular IP – Data Communication



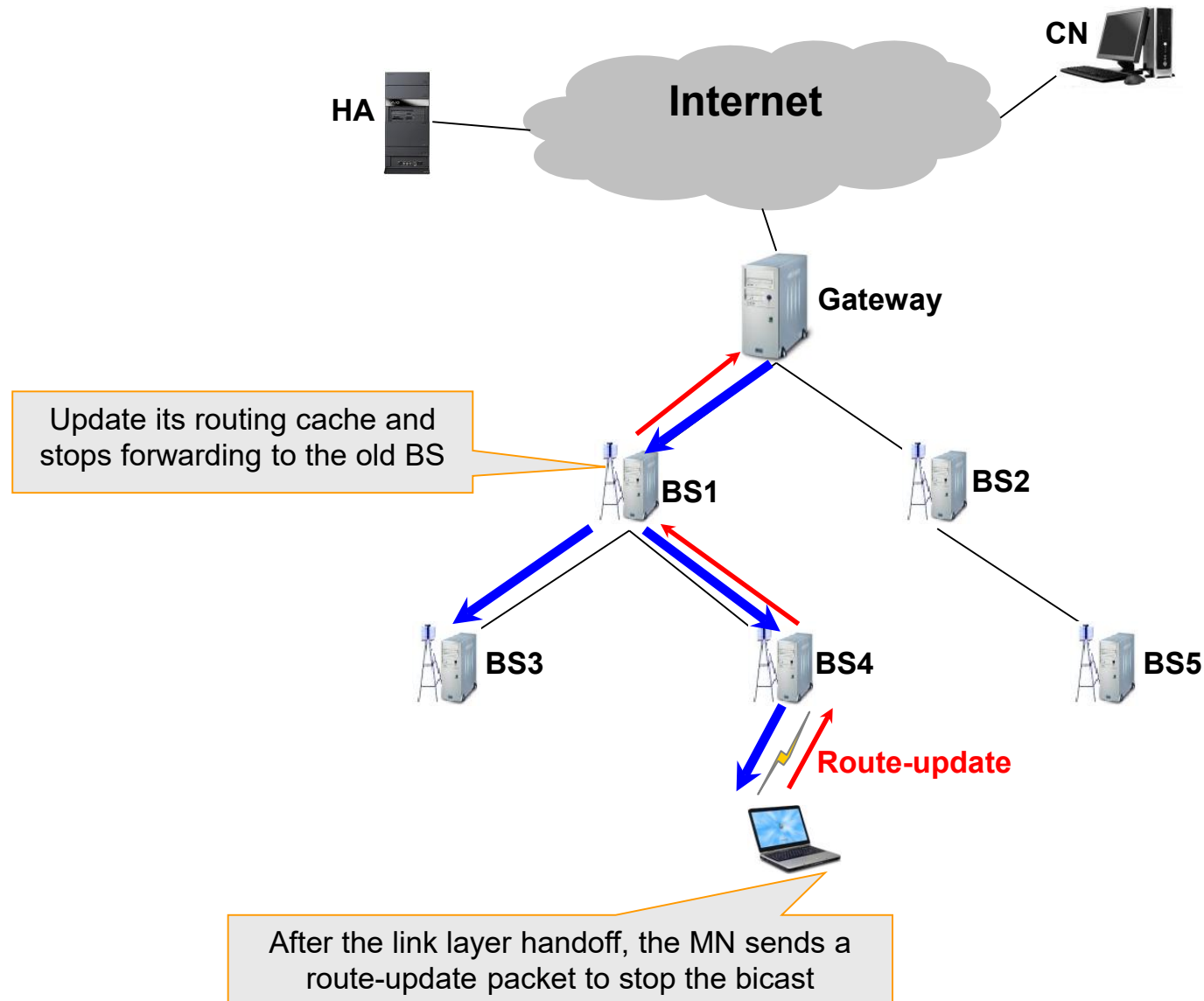
Cellular IP – Hard Handoff



Cellular IP – Semi-Soft Handoff



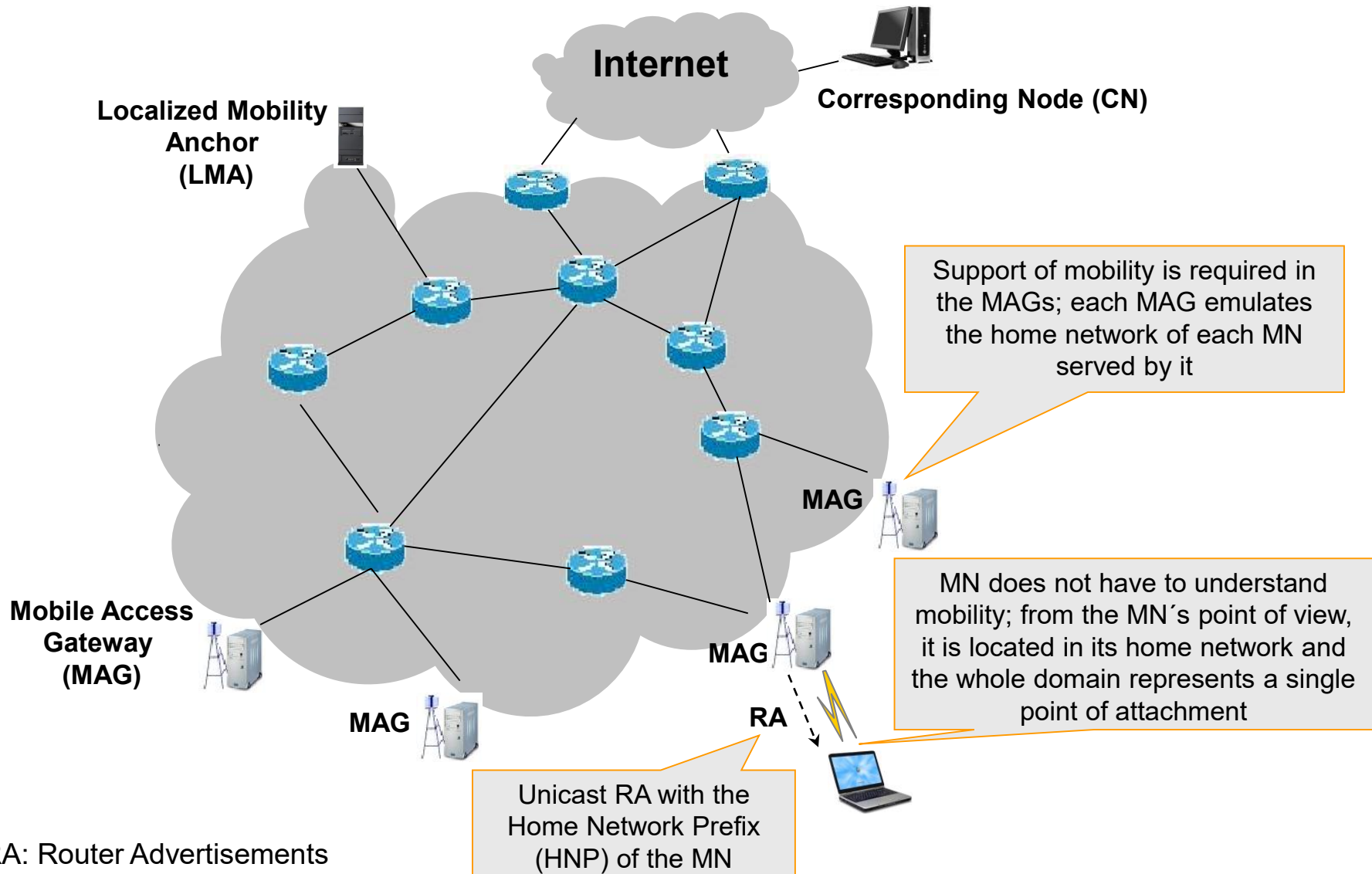
Cellular IP – Semi-Soft Handoff



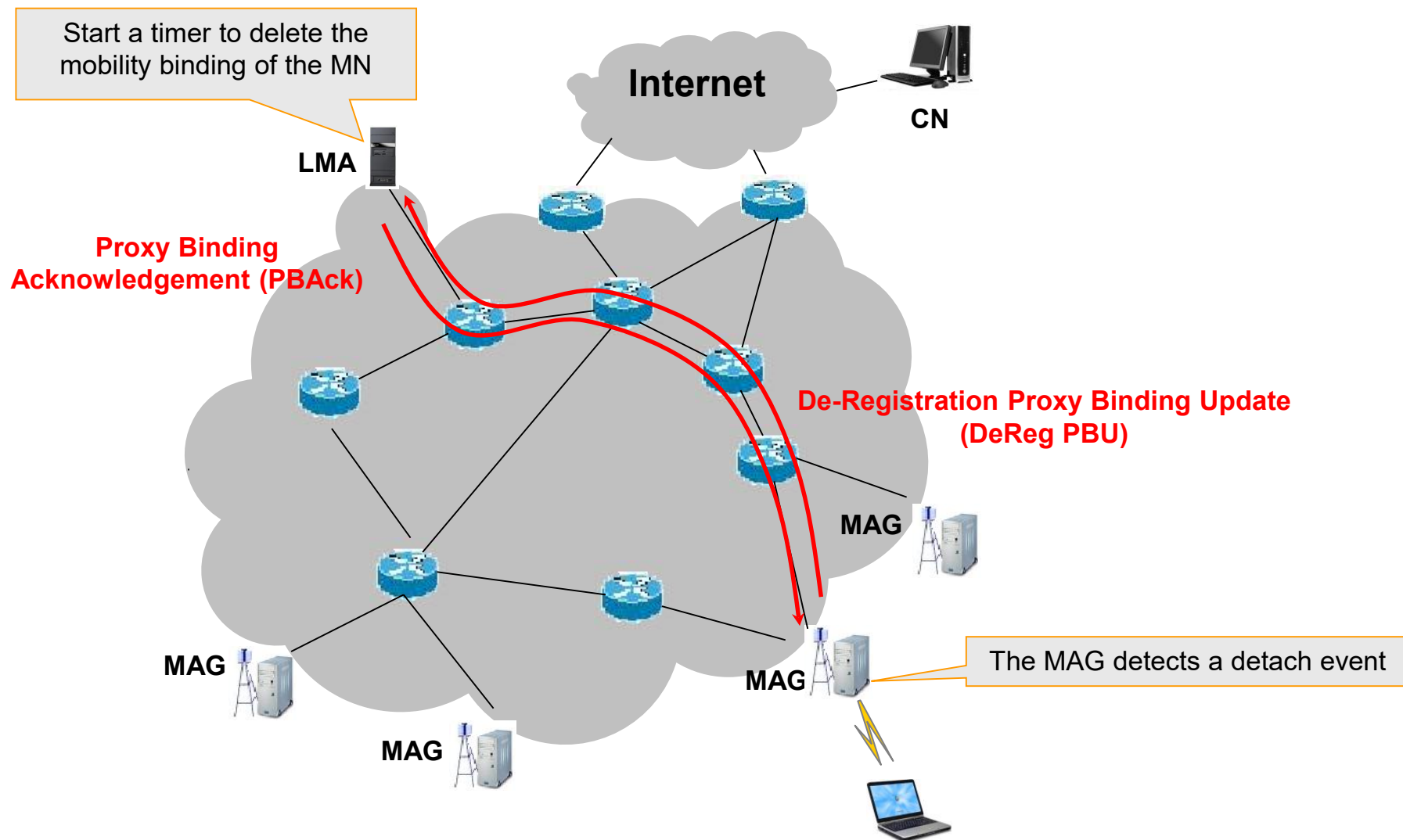
Cellular IP – Pros & Cons

- Pros
 - Localizing of mobility management inside administrative domains
 - Improving the performance: reduced handoff latency, less lost packets, etc.
 - Reducing the signaling towards the HA
 - Mobile to mobile communication within the domain has to be routed via the gateway
- Cons
 - Every node in the domain has to be mobility-aware
 - Restrictions on the network topology
 - Single point of failure (all traffic travels through the gateway)
 - Signaling and packet forwarding overhead inside the domain
 - Reduced TCP throughput due to new layer 3 dynamic routing protocol
 - Security issues: authentication of arriving MNs, authentication of route-update messages, etc.

Network-based MM: Proxy MIPv6

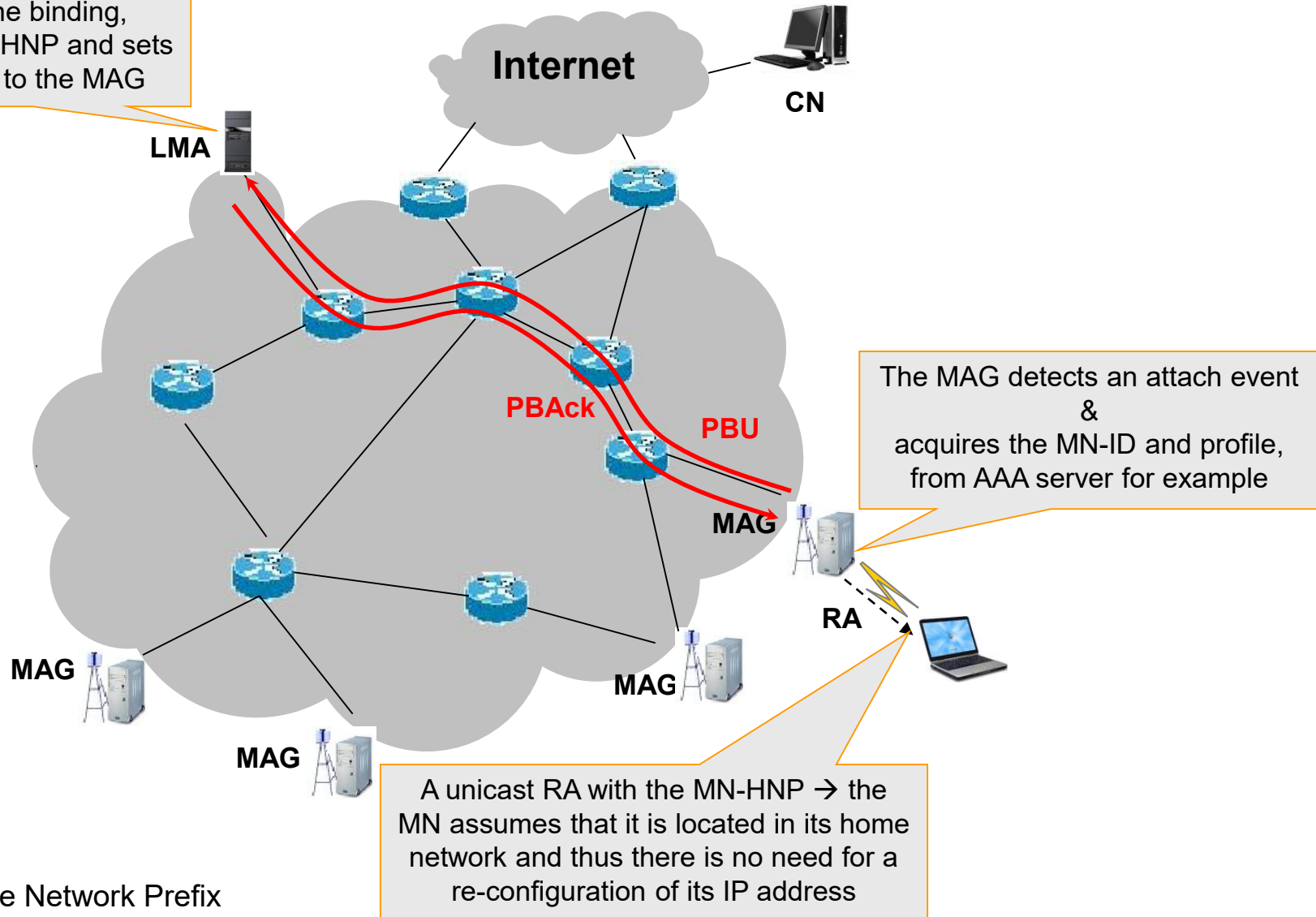


Proxy MIPv6 – Handoff



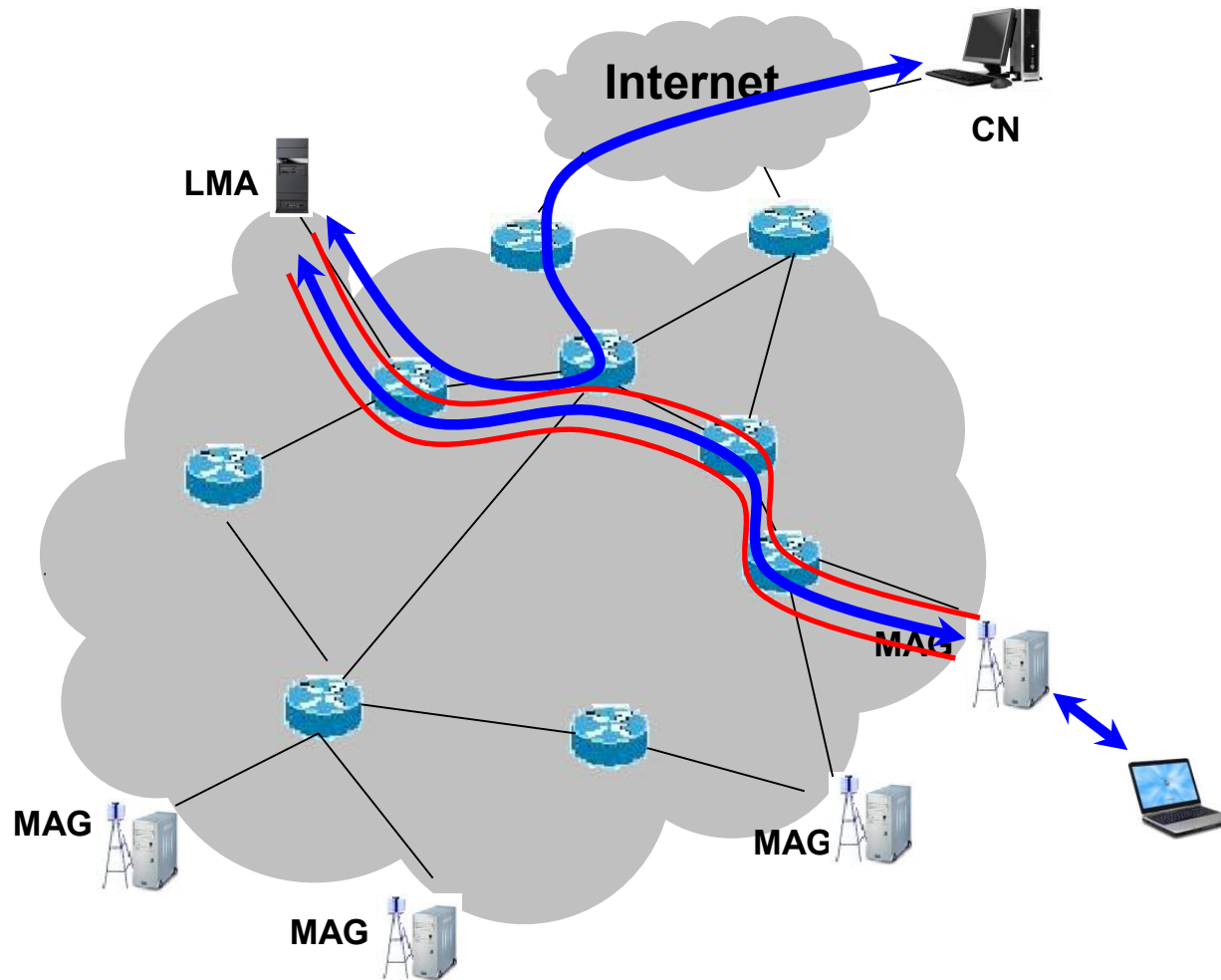
Proxy MIPv6 – Handoff

Accepts the binding,
allocates MN-HNP and sets
up a tunnel to the MAG



HNP: Home Network Prefix

Proxy MIPv6 – Data Communication



Proxy MIPv6 – Pros & Cons

- Pros
 - Transparency to MNs (MNs with no mobility support can be mobile)
 - Protocol is robust against control messages dropping (no control messages are sent on wireless links)
 - Reduced signaling cost
 - Terminating the tunnel in the MAG instead of MNs reduces the data traffic volume sent over the wireless link
- Cons
 - Triangular routing → increased end-to-end delay
 - Overhead due to the encapsulation
 - Handoff latency due to contacting the LMA each time the MN moves from one point of attachment to another → communication disruption
 - Route optimization is not possible

Conclusions

- Mobility can be supported by different layers of the TCP/IP model
- Network layer mobility approaches are the most deployed solutions
 - **Terminal-based** approaches interact with MNs to support mobility
 - **Network-based** approaches do not involve MNs in mobility support
- The approaches are categorized into macro and micro mobility management approaches
 - **Macro mobility** approaches aim at supporting global mobility
 - **Micro mobility** approaches aim at accelerating the mobility management through processing of mobility locally

References

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- Ramjee Prasad, Marina Ruggieri: Technology Trends in Wireless Communications, Artech House, 2003
- Wisely, Eardley, Burness: IP for 3G: Networking Technologies for Mobile Communications, Wiley, 2002 (chapter 5 provides classification of mobility mechanisms and overview on protocols, including session-layer mobility)
- D. Johnson, C. Perkins, J. Arkko, “Mobility Support in IPv6”, RFC 3775, June 2004.
- S. Gundavelli, et.al., „Proxy Mobile IPv6“, RFC 5213, Aug. 2008.
- J. Wozniak, Mobility management solutions for current IP and future networks, Telecommunication Systems, 61(2), Springer, pp 257–275, February 2016.
- Important standards:
 - Mobile IP: RFC 3220, formerly: RFC 2002
 - Reverse tunneling: RFC 3024, formerly: 2344
 - DHCP: RFC 2131, RFC 2132
 - ICMP: RFC 1256